

From

The Students of
B.Tech Computer Science and Engineering
Semester IV
Government Engineering College, Thrissur

To

The Head of the Department
Department of Computer Science and Engineering
Government Engineering College, Thrissur

Date: _____

Subject: Request for approval of MOOC courses as per KTU B.Tech Regulations

Respected Sir/Madam,

We, the students of B.Tech Computer Science and Engineering, Semester IV, Government Engineering College, Thrissur, respectfully submit this request seeking approval for the MOOC courses proposed by us, in accordance with the provisions of the APJ Abdul Kalam Technological University (KTU) B.Tech Regulations.

We propose to undertake the attached MOOC courses as equivalents to the corresponding subjects prescribed in the KTU curriculum, subject to the approval of the competent authority.

For each proposed MOOC course, the following documents are enclosed for your kind consideration:

- The KTU syllabus of the course to be replaced
- The MOOC syllabus / course outline
- Details of the offering institute/university, course duration, credit information, and assessment structure

We assure that we shall strictly adhere to all guidelines prescribed by the college and the university, complete the approved MOOC courses within the stipulated time, and submit the required course completion certificates and supporting documents within the prescribed deadlines.

We kindly request the Department Academic Committee and the competent authority to review the enclosed documents and grant approval for the proposed MOOC courses.

Thanking you.

Yours faithfully,

Students of

B.Tech Computer Science and Engineering
Semester IV
Government Engineering College, Thrissur

INDEX

S.No	KTU Course	MOOC Course	Start Page
1	PECST745 - Computer Vision	Computer Vision and Image Processing - Fundamentals and Applications	3
2	PECST747 - Blockchain and Cryptocurrencies	Blockchain and its Applications	10
3	PECST749 - Approximation Algorithms	APPROXIMATION ALGORITHM	17
4	PECST785 - Algorithms For Data Science	Data Science for Engineers	23
5	PECST759 - Parallel Algorithms	Parallel Computer Architecture	30
6	PECST753 - Fuzzy Systems	FUZZY SETS, LOGIC AND SYSTEMS & APPLICATIONS	36
7	PECST757 - High Performance Computing	High Performance Scientific Computing	42
8	PECST759 - Parallel Algorithms	Parallel Algorithms	48
9	OEECT723 - Optimization Techniques	Optimization from fundamentals	54
10	OEMET722 - Robotics	Robotics and Control : Theory and Practice	60
11	PECST862 - Natural Language Processing	Natural Language Processing	68
12	OEERT834 - CLOUD COMPUTING AND APPLICATIONS	Cloud Computing	74
13	OEEET832 - PLC & Automation	Industrial Automation and Control	80
14	OEECT831 - Internet of Things	Introduction to Internet Of Things	87
15	UEHUT803 - Organizational Behavior and Business Communication	Organizational Behaviour: Individual Dynamics in Organization	92
16	UEHUT704 - Project Management: Planning, Execution, Evaluation and Control	Project Management : Planning, Execution, Evaluation And Control	100
17	UEHUT704 - Project Management: Planning, Execution, Evaluation and Control	Project Management	106
18	HNCST709 - Advanced Cryptography	Cryptography and Network Security	113
19	HNCST509 - Object Oriented Design using UML	Object Oriented System Development using UML, Java And Patterns	118
20	HNCST609 - Advanced Algorithms	Design and Analysis of Algorithms	123
21	PECST742 - Web Programming	Web Technology	129

MOOC APPROVAL PROPOSAL

KTU Course Name	Computer Vision
Course Category	Program Elective 4
Course Code	PECST745
MOOC Course Name	Computer Vision and Image Processing - Fundamentals and Applications
NPTEL Subject ID	108103174
Course ID	noc26-ee31
Course URL	https://nptel.ac.in/courses/108103174
COORDINATORS	Prof. M. K. Bhuyan
Offering Institute	IIT Guwahati
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

COMPUTER VISION

Course Code	PECST745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To cover the basics of image formation, key computer vision concepts, methods, techniques, pattern recognition, and various problems in designing computer vision and object recognition systems.
2. To enable the learners to understand the fundamentals of computer vision and machine learning models to develop applications in computer vision.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals in Computer Vision :- Camera Calibration- Pinhole camera model, Geometric Image Features - Curves, Surfaces, Analytical Image Features - Elements of Analytical Euclidean Geometry, Geometric Camera Parameters, Stereopsis - Binocular Camera Geometry, Epipolar Constraint, Binocular Reconstruction, Local Methods for Binocular Fusion, Global Methods for Binocular Fusion.	9
2	Features and Filters :- Linear Filters- Linear Filters and Convolution, Shift Invariant Linear Systems. Estimating Derivatives with Finite Differences, Noise, Edges and Gradient-based Edge Detectors Image Gradients - Computing the Image Gradient, Gradient Based Edge and Corner Detection. Filters as Templates - Normalized Correlation and Finding Patterns.	9

3	Machine Learning for Computer Vision :- Machine Learning - Introduction, Dataset for Machine Perception- Labelled and Unlabelled Data, Basics of Classification and Clustering, Multi-Class Perspective. Machine Learning for Computer Vision -Machine Learning -Deep Learning Use Cases. Machine Learning Models for Vision - Image Vision-Pretrained Model, Transfer Learning, Fine-Tuning, Convolutional Networks, Convolutional Filters, Stacking Convolutional Layers, Pooling Layers - AlexNet, VGG19, , Modular architecture - ResNet, Neural Architecture Search Design - NASNet	9
4	Segmentation and Object detection :- Segmentation Using Clustering Methods - Human vision- Grouping and Gestalt, Applications- Shot Boundary Detection, Background Subtraction, Image Segmentation by Clustering Pixels- Simple Clustering Methods, Clustering and Segmentation by K-means Object detection - YOLO, Segmentation-Mask R-CNN and Instance Segmentation, U-Net and Semantic Segmentation, Model Quality Metrics <i>A case study to compare performance of various models on a suitable dataset.</i>	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts and terminologies like Camera Calibration, Stereopsis in computer vision	K2
CO2	Apply filters for feature extraction and for finding patterns.	K3
CO3	Build different machine learning models for computer vision	K3
CO4	Implement segmentation and object detection models	K3
CO5	Analyze different machine learning models for segmentation/object detection.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3									3
CO3	3	3	3									3
CO4	3	3	3	3								3
CO5	3	3	3	3	3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer vision: A modern approach	Forsyth, David, and Jean Ponce	Prentice hall	2011
2	Emerging topics in computer vision	Medioni, Gerard and Sing Bing Kang	PHI	2004
3	Practical Machine Learning for Computer Vision	Valliappa Lakshmanan, Martin Görner, Ryan Gillard	O'Reilly Media	2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer vision: algorithms and applications	Szeliski, Richard	Springer Science & Business Media	2010
2	Image Segmentation: Principles, Techniques, and Applications	Tao Lei, Asoke K. Nandi	John Wiley & Sons	2022
3	Deep Learning in Computer Vision Principles and Applications	Ali Ismail Awad, Mahmoud Hassaballah	CRC Press	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Computer Vision and Image Processing - Fundamentals and Applications by Prof. M. K. Bhuyan at IIT Guwahati https://onlinecourses.nptel.ac.in/noc23_ee39/preview
2	Computer Vision by Prof. Jayanta Mukhopadhyay at IIT Kharagpur https://onlinecourses.nptel.ac.in/noc19_cs58/preview
3	
4	Deep Learning for Computer Vision by Prof. Vineeth N Balasubramanian at IIT Hyderabad https://onlinecourses.nptel.ac.in/noc21_cs93/preview
	COVID-Net Open Source Initiative - COVIDx CT-3 Dataset https://www.kaggle.com/datasets/hgunraj/covidxct



COMPUTER VISION AND IMAGE PROCESSING - FUNDAMENTALS AND APPLICATIONS

PROF. M.K. BHUYAN

Department of Electrical Engineering
IIT Guwahati

PRE-REQUISITES : Basic co-ordinate geometry, matrix algebra, linear algebra and random process.

INTENDED AUDIENCE : UG, PG and Ph.D students

INDUSTRIES APPLICABLE TO : The software industries that develop computer visions apps would be benefitted from this course.

COURSE OUTLINE :

The intent of this course is to familiarize the students to explain the fundamental concepts/issues of Computer Vision and Image Processing, and major approaches that address them. This course provides an introduction to computer vision including image acquisition and image formation models, radiometric models of image formation, image formation in the camera, image processing concepts, concept of feature extraction and selection for pattern classification/recognition, and advanced concepts like motion estimation and tracking, image classification, scene understanding, object classification and tracking, image fusion, and image registration, etc.

This course will cover the fundamentals of Computer Vision. It is suited for mainly students who are interested in doing research in the area of Computer Vision. After completing the course, the students may expect to have the knowledge needed to read and understand more advanced topics and current research literature, and the ability to start working in industry or in academic research in the field of Computer Vision and Image Processing. They can also apply all these concepts for solving the real-world problems.

ABOUT INSTRUCTOR :

Prof. Manas Kamal Bhuyan received a Ph.D. degree in electronics and communication engineering from the India Institute of Technology (IIT) Guwahati, India. He was with the School of Information Technology and Electrical Engineering, University of Queensland, St. Lucia, QLD, Australia, where he was involved in postdoctoral research. Subsequently, he was a Researcher with the SAFE Sensor Research Group, NICTA, Brisbane, QLD, Australia. He was an Assistant Professor with the Department of Electrical Engineering, IIT Roorkee, India and Jorhat Engineering College, Assam, India. In 2014, he was a Visiting Professor with Indiana University and Purdue University, Indiana, USA. Dr. Bhuyan was a recipient of the National Award for Best Applied Research/ Technological Innovation, which was presented by the Honorable President of India, the Prestigious Fulbright-Nehru Academic and Professional Excellence Fellowship, and the BOYSCAST Fellowship. He is an IEEE senior member. He is currently a Professor with the Department of Electronics and Electrical Engineering, IIT Guwahati, and Associate Dean of Infrastructure, Planning and Management, IIT Guwahati. His current research interests include image/video processing, computer vision, machine learning and human computer interactions (HCI), virtual reality and augmented reality. He has almost 25 years of industry, teaching, and research experience. He is the author of the book text book "Computer Vision and Image Processing: Fundamentals and Applications". For more details www.iitg.ac.in/mkb

COURSE PLAN :

Week 1: Introduction to Computer Vision and Basic Concepts of Image Formation: Introduction and Goals of Computer Vision and Image Processing, Image Formation Concepts.

Week 2: Fundamental Concepts of Image Formation: Radiometry, Geometric Transformations, Geometric Camera Models.

Week 3: Fundamental Concepts of Image Formation: Camera Calibration, Image Formation in a Stereo Vision Setup, Image Reconstruction from a Series of Projections.

Week 4: Image Processing Concepts: Image Transforms.

Week 5: Image Processing Concepts: Image Transforms, Image Enhancement.

Week 6: Image Processing Concepts: Image Filtering, Colour Image Processing, Image Segmentation

Week 7: Image Descriptors and Features: Texture Descriptors, Colour Features, Edges/Boundaries.

Week 8: Image Descriptors and Features: Object Boundary and Shape Representations.

Week 9: Image Descriptors and Features: Interest or Corner Point Detectors, Histogram of Oriented Gradients, Scale Invariant Feature Transform, Speeded up Robust Features, Saliency

Week 10: Fundamentals of Machine Learning: Linear Regression, Basic Concepts of Decision Functions, Elementary Statistical Decision Theory, Parameter Estimation, Clustering for Knowledge Representation, Dimension Reduction, Linear Discriminant Analysis.

Week 11: Applications of Computer Vision: Artificial Neural Network for Pattern Classification, Convolutional Neural Networks, Autoencoders.

Week 12: Applications of Computer Vision: Gesture Recognition, Motion Estimation and Object Tracking, Programming Assignments.

MOOC APPROVAL PROPOSAL

KTU Course Name	Blockchain and Cryptocurrencies
Course Category	Program Elective 4
Course Code	PECST747
MOOC Course Name	Blockchain and its Applications
NPTEL Subject ID	106105235
Course ID	noc26-cs34
Course URL	https://nptel.ac.in/courses/106105235
COORDINATORS	Prof. Sandip Chakraborty,; Prof. Shamik Sural
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

BLOCKCHAIN AND CRYPTOCURRENCIES

Course Code	PECST747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBCST604	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of blockchain architecture, elements, types (public, private, consortium), and industry applications.
2. To help the learners to assess strengths and weaknesses of various blockchain consensus mechanisms (e.g., Proof of Work, Proof of Stake, Practical Byzantine Fault Tolerance).
3. To enable learners to use blockchain real-world applications in government, healthcare, finance, and supply chain management, identifying implementation opportunities and challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Blockchain Fundamentals Introduction, Blockchain Definition, Deciphering the Blockchain, Features and challenges of Blockchain, Applications in Blockchain, Decentralisation, Distributed Ledger Technology, Blockchain variants.	7
2	Cryptography in Blockchain and Consensus Mechanisms Concept of Hashing, Creating a Transaction Hash, Merkle Trees - Importance of Merkle tree, Chaining of Blocks, Building the Network, Accessing the network, Types of Wallets. Need for Consensus, Two Generals' Problem, Byzantine Generals' Problem, Byzantine Fault Tolerance (BFT), Practical Byzantine Fault Tolerance (PBFT)- working, Paxos and Raft Algorithms.	9
3	Cryptocurrencies - Bitcoin and Ethereum Bitcoin: Components, Nodes in Bitcoin network, Transactions and memory	10

	pools, Proof of Work-Mining Cryptocurrencies, Hard and Soft Forks, Tracking Bitcoins-Unspent Transaction Outputs. Ethereum: Transition from Bitcoin to Ethereum, Concept of Ethereum World Computer, Ethereum Virtual Machine, Ethereum Network, Transition from PoW to PoS- Working of PoS, Smart Contracts in Ethereum, Decentralised Applications in Ethereum, Tools used in Ethereum.	
4	Blockchain Ethereum Platform using Solidity and Use Cases in Blockchain :- Solidity Language - Remix IDE, Structure of a Smart Contract Program, Modifiers, Events, Functions, Inheritance, External Libraries, Error Handling. Permissioned Blockchains, Introduction to Hyperledger Foundation, Hyperledger Distributed Ledger frameworks, Hyperledger Fabric. Use Cases in Blockchain - Finance, Education, Government, Healthcare and Supply Chain Management.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of Blockchain technology.	K2
CO2	Illustrate the cryptographic building blocks of Blockchain technology and understand the consensus mechanisms.	K2
CO3	Explain the concepts of cryptocurrency bitcoin, mining processes, and wallet management.	K2
CO4	Use the concepts of Ethereum platform and understand the use cases of blockchain technology	K3
CO5	Develop skills in designing and deploying simple applications using Solidity language.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									2
CO2	3	3	3									2
CO3	3	3	3									2
CO4	3	3	3		3							2
CO5	3	3	3	3	3							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Blockchain Technology: Algorithms and Applications	Asharaf S, Sivadas Neelima, Adarsh S, Franklin John	Wiley	1/e, 2023
2	Blockchain Technology	Chandramauoli Subrahmaniyan, Asha A George	Universities Press.	1/e, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Blockchain Technology - Concepts and Applications.	Kumar Saurabh, Ashutosh Saxena	Wiley	1/e, 2020
2	Mastering Blockchain	Imran Bashir	Packt Publishing	1/e, 2020
3	Solidity programming Essentials: A beginner's guide to build smart contracts for Ethereum and blockchain	Ritesh Modi	Packt Publishing	1/e, 2018.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLrKK422S1aMma8lDA2JJjEUUpC2ycuApuC&si=1OXYDEZ4A5M8M4Q
2	https://youtube.com/playlist?list=PLHRLZtgrF2jl8yqucJsMFqh5XpRLTgCI4
3	https://youtube.com/playlist?list=PL6gx4Cwl9DGBrtyuJUiv9Lq5CAYpN8Gl
4	https://youtube.com/playlist?list=PLWUCKsxdKl0oksYr6lG_wRsaSUySQC0ck



BLOCKCHAIN AND ITS APPLICATIONS

PROF. SANDIP CHAKRABORTY

Department of Computer Science and Engineering
IIT Kharagpur

PROF. SHAMIK SURAL

Department of Computer Science and Engineering
IIT Kharagpur

PRE-REQUISITES : Computer Networks; Operating Systems; Cryptography and Network Security

INTENDED AUDIENCE : Undergraduate Students, Postgraduate Students, Industry Associates

INDUSTRIES APPLICABLE TO : IBM; HPE; Intel; Any startups working on Blockchain

COURSE OUTLINE :

In the last few years, Blockchain technology has generated massive interest among governments, enterprises, and academics, because of its capability of providing a transparent, secured, tamper-proof solution for interconnecting different stakeholders in a trustless setup. In January 2021, the Ministry of Electronics and Information Technology (MeiTY), Government of India, published the first draft of the "National Strategy on Blockchain" that highlights 17 potential applications that are of national interest. Against this backdrop, this subject will cover the basic design principles of Blockchain technology and its applications over different sectors. Additionally, the course also provides tutorials on setting up blockchain applications using one of the well-adopted permissionless blockchain platforms -Ethereum, and one permissioned blockchain platform - Hyperledger.

ABOUT INSTRUCTOR :

Prof. Sandip Chakraborty is working as an Associate Professor in the Department of Computer Science and Engineering at the Indian Institute of Technology (IIT) Kharagpur. He obtained his Bachelor of Engineering (BE) degree from Jadavpur University, Kolkata in 2009 and Master of Technology (M Tech) and Doctor of Philosophy (Ph.D.), both from IIT Guwahati, in 2011 and 2014, respectively. The primary research interests of Dr. Chakraborty is in the intersection of Computer Systems, Pervasive Computing, and Human-Computer Interaction. Dr. Chakraborty is leading the System and Mobile Research Lab at IIT Kharagpur, focusing on various aspects of computer systems and networks along with the design and development of ubiquitous and pervasive sensing systems. He is one of the founding members of ACM IMOBILE, the ACM SIGMOBILE chapter in India. He is working as an Area Editor of Elsevier Ad Hoc Networks and Elsevier Pervasive and Mobile Computing journal. He has received various awards and accolades including Excellent Young Teacher Award 2021, INAE Young Engineers' Award, Fellow of National Internet Exchange of India (NIXI), and so on. Further details about his works and publications can be obtained from <https://cse.iitkgp.ac.in/~sandipc/index.html>.

Prof. Shamik Sural is a full professor in the Department of Computer Science and Engineering, Indian Institute of Technology (IIT) Kharagpur. He received the Ph.D. degree from Jadavpur University, Kolkata, India in the year 2000. Before joining IIT Kharagpur in 2002, he spent more than a decade in the Information Technology industry working in India as well as in Michigan, USA. Prof. Sural was a recipient of the Alexander van Humboldt Fellowship for Experienced Researchers in 2009, which enabled him to carry out collaborative research at the Technical University of Munich, Germany. Later in 2018, he was awarded a Humboldt Foundation Alumni Fellowship for a renewed research stay in Germany. He also spent the Fall 2019 semester at Rutgers University, USA as a Fulbright scholar engaged in both teaching and research. During this period, he visited several other universities in the USA for delivering invited lectures. Prof. Sural is a senior member of IEEE and has previously served as the Chairman of the IEEE Kharagpur Section in 2006. He is currently on the editorial boards of IEEE Transactions on Dependable & Secure Computing, IEEE Transactions on Services Computing and Sadhana – a journal of the Indian Academy of Sciences. He has published more than two hundred research papers in reputed international journals and conferences. His research interests include computer security, data mining and multimedia systems.

COURSE PLAN :

Week 1: Introduction to Blockchain Technology and its Importance

Week 2: Basic Crypto Primitives I - Cryptographic Hash

Week 3: Basic Crypto Primitives II - Digital Signature

Week 4: Evolution of the Blockchain Technology

Week 5: Elements of a Blockchain

Week 6: Blockchain Consensus I - Permissionless Models

Week 7: Blockchain Consensus II - Permissioned Models

Week 8: Smart Contract Hands On I - Ethereum Smart Contracts (Permissionless Model)

Week 9: Smart Contract Hand On II - Hyperledger Fabric (Permissioned Model)

Week 10: Decentralized Identity Management

Week 11: Blockchain Interoperability

Week 12: Blockchain Applications

MOOC APPROVAL PROPOSAL

KTU Course Name	Approximation Algorithms
Course Category	Program Elective 4
Course Code	PECST749
MOOC Course Name	APPROXIMATION ALGORITHM
NPTEL Subject ID	106105471
Course URL	https://nptel.ac.in/courses/106105471
COORDINATORS	Prof. Palash Dey
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

APPROXIMATION ALGORITHMS

Course Code	PECST749	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To provide a deep understanding of approximation algorithms, including their design, analysis, and application to various optimization problems.
2. To equip the skills to evaluate and analyze the efficiency and effectiveness of approximation techniques. This includes understanding performance metrics, approximation ratios, and the theoretical limits of approximation algorithms, as well as applying these techniques to complex problems in network design, combinatorial optimization, and other areas.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Approximation Algorithms - Introduction to approximation algorithms, Performance guarantees: approximation ratio and factor, Examples of approximation problems. (Chapter 1) Greedy Algorithms - Introduction to greedy algorithms, Set cover problem, Vertex cover problem. (Chapter 2) Local Search Algorithms - Local search techniques, k-Median and k-Center problems, Analysis of local search algorithms. (Chapter 3)	9
2	Linear Programming Relaxation - Introduction to linear programming (LP), LP relaxation of combinatorial problems, Primal-dual method. (Chapter 4) Rounding Techniques - Randomized rounding, Deterministic rounding, Applications to various problems. (Chapter 5) Integer Programming and Cutting Planes - Integer programming formulation, Cutting plane methods, Applications in network design. (Chapter 6)	9
3	Semi-Definite Programming - Introduction to semi-definite programming (SDP), Goemans-Williamson algorithm for MAX-CUT, Other applications of SDP. (Chapter 8) Approximation Schemes - Polynomial-time approximation schemes (PTAS),	9

	Fully polynomial-time approximation schemes (FPTAS), Examples: knapsack problem, Euclidean TSP. (Chapter 9)	
4	Inapproximability Results - Introduction to inapproximability, Reductions and hardness of approximation, PCP theorem and its implications. (Chapter 10) Network Design Problems - Steiner tree problem, Traveling Salesman Problem (TSP), Multicommodity flow problem. (Chapter 7)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate a foundational understanding of approximation algorithms, including performance guarantees, approximation ratios, and common examples of approximation problems.	K3
CO2	Illustrate the principles of greedy algorithms and apply them to solve classic problems such as the set cover and vertex cover problems, understanding their efficiency and limitations.	K3
CO3	Show proficiency in local search algorithms and linear programming relaxation methods, including the primal-dual method, and apply these techniques to solve combinatorial optimization problems.	K3
CO4	Understand and implement rounding techniques, both randomized and deterministic, and learn the basics of semi-definite programming (SDP), including algorithms like Goemans-Williamson for the MAX-CUT problem.	K3
CO5	Demonstrate polynomial-time approximation schemes (PTAS) and fully polynomial-time approximation schemes (FPTAS), and explore inapproximability results, including reductions, hardness of approximation, and the PCP theorem.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								2
CO2	3	3	3	3								2
CO3	3	3	3	3								2
CO4	3	3	3	3								2
CO5	3	3	3	3								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Approximation Algorithms	Vijay V. Vazirani	Springer Nature (SIE)	2/e, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The design of approximation algorithms	David Williamson and David Shmoys	Cambridge University Press	1/e, 2011
2	Randomized Algorithms	Rajeev Motwani and Prabhakar Raghavan	Cambridge University Press	1/e, 2004
3	Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis	Michael Mitzenmacher and Eli Upfal	Cambridge University Press	3/e, 2017
4	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	The MIT Press	4/e, 2023
5	The Probabilistic Method	Noga Alon and Joel H. Spencer	Wiley-Blackwell	4/e, 2016
6	Computational Complexity: A Modern Approach	Sanjeev Arora and Boaz Barak	Cambridge University Press	1/e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106105471
2	https://nptel.ac.in/courses/106105471
3	https://nptel.ac.in/courses/106105471
4	https://nptel.ac.in/courses/106105471



APPROXIMATION ALGORITHM

PROF. PALASH DEY

Department of Computer Science and Engineering
IIT Kharagpur

PRE-REQUISITES : Knowledge of algorithm design principles.

INTENDED AUDIENCE : Under-graduate and post-graduate students and faculty members.

INDUSTRY SUPPORT : All software companies especially Google, Microsoft, etc.

COURSE OUTLINE :

Many real-world problems are NP-complete. Hence, they are unlikely to admit a polynomial-time algorithm. In this course, we will study various techniques to design efficient algorithms to compute an approximately optimal solutions.

ABOUT INSTRUCTOR :

Prof. Palash Dey is an Assistant Professor in the Department of Computer Science and Engineering at Indian Institute of Technology, Kharagpur since 2018. Before joining IIT Kharagpur, he was a post-doctoral INSPIRE faculty in TIFR, Mumbai for one year. He finished his Ph.D. and M.E. from the Department of Computer Science and Automation at Indian Institute of Science, Bangalore in 2017 and 2013 respectively. Prior to that, he finished his B.E. from the Department of Computer Science and Engineering at Jadavpur University in 2010. His primary field of research is algorithmic game theory. He is broadly interested in theoretical computer science.

COURSE PLAN :

Week 1: Review of NP-Completeness

Week 2: Approximation algorithm for the vertex cover, set cover, and traveling salesman problem

Week 3: Approximation algorithm for the Knapsack problem and the job scheduling problems

Week 4: Basics of linear programming

Week 5: Deterministic rounding: set cover, vertex cover problems

Week 6: Deterministic rounding: steiner tree, facility location problems

Week 7: Randomized rounding: max-SAT problem

Week 8: Randomized rounding: steiner tree, facility location problems

Week 9: Primal-dual method: set cover

Week 10: Primal-dual method: steiner tree

Week 11: Semidefinite programming: max cut

Week 12: Hardness of approximation: travelling salesman problem, job scheduling

MOOC APPROVAL PROPOSAL

KTU Course Name	Algorithms For Data Science
Course Category	Program Elective 5
Course Code	PECST785
MOOC Course Name	Data Science for Engineers
NPTEL Subject ID	106106179
Course ID	noc26-cs65
Course URL	https://nptel.ac.in/courses/106106179
COORDINATORS	Prof. Shankar Narasimhan; Prof. Ragunathan Rengaswamy
Offering Institute	IIT Madras
Duration	8 weeks
Platform	NPTEL

SEMESTER S7
ALGORITHMS FOR DATA SCIENCE
(Common to CS/AM/CM)

Course Code	PECST785	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Mins.
Prerequisites (if any)	PCCST303 PCCST502	Course Type	Theory

Course Objectives:

1. To equip students with the ability to design, analyze, and implement advanced algorithms that are fundamental to data science, enabling them to process and analyze large-scale datasets efficiently and effectively.
2. To provide hands-on experience through real-world projects that require students to apply algorithmic techniques to solve data science problems, strengthen the development of practical skills in data manipulation, analysis, and interpretation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Foundations of Data Science Algorithms Introduction to Data Science and Algorithms - Overview of data science and its significance, Role of algorithms in data science; Data Preprocessing Techniques - Data cleaning, transformation, and normalization, Handling missing data, outliers, and data imputation techniques; Dimensionality reduction techniques - Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE); Algorithmic Approaches to Data Sampling - Random sampling, stratified sampling, and bootstrapping, Importance of representative sampling in data analysis. Project 1: Data Cleaning and Preprocessing - Develop a pipeline for cleaning and preprocessing a large, messy dataset like UCI Machine Learning Repository - Adult Data Set Tasks: Handle missing values, outliers, and noisy data. Apply	9

	dimensionality reduction techniques to simplify the dataset. Implement data transformation and normalization processes.	
2	Algorithms for Data Summarization and Visualization :- Data Summarization Techniques - Central tendency measures: mean, median, mode; Dispersion measures - variance, standard deviation, Interquartile range (IQR), Quantiles, percentiles, and outlier detection; Visualization Algorithms - Basics of data visualization, histograms, bar charts, scatter plots; Advanced visualization techniques - heatmaps, correlation matrices, and pair plots; Visualization tools and libraries - Matplotlib, Seaborn, Plotly; Algorithmic Approaches to Data Grouping - Clustering: k-means, hierarchical clustering, DBSCAN; Association rule learning - Apriori, FP-Growth. Project 2: Exploratory Data Analysis and Visualization Perform exploratory data analysis (EDA) and create visualizations to uncover patterns and insights in the dataset like Kaggle - Titanic Dataset Tasks: Summarize the dataset using statistical measures. Create various visualizations to explore relationships and patterns in the data. Implement clustering algorithms to identify natural groupings within the data.	9
3	Algorithms for Data Modeling :- Regression Algorithms - Linear regression and polynomial regression; Regularization techniques - Ridge, Lasso, Elastic Net; Evaluation metrics - RMSE, MAE, R²; Classification Algorithms - Logistic regression, decision trees, and k-Nearest Neighbors (k-NN); Performance metrics - accuracy, precision, recall, F1-score, ROC-AUC; Algorithmic Optimization Techniques - Gradient descent and its variants: stochastic, mini-batch; Hyperparameter tuning - grid search, random search, Bayesian optimization. Project 3: Predictive Modeling and Evaluation - Build and evaluate predictive models using regression and classification algorithms using datasets like Kaggle - House Prices: Advanced Regression Techniques Tasks: Implement linear and polynomial regression models to predict house prices. Apply classification algorithms to classify houses into different categories. Evaluate the models using appropriate performance metrics and fine-tune them for better accuracy.	9
4	Algorithms for Big Data and Scalability :- Introduction to Big Data Algorithms - Overview of big data challenges and	9

	processing techniques; Distributed computing frameworks - Hadoop, Spark; MapReduce paradigm - concepts and applications; Scalable Data Processing Algorithms - Algorithms for large-scale data processing : sorting, searching, filtering; Data partitioning and shuffling techniques in distributed systems; Handling data with memory constraints - external memory algorithms. Project 4: Scalable Data Processing with Spark - Implement scalable algorithms using Apache Spark to process large datasets efficiently using datasets like Kaggle - Google Analytics Customer Revenue Prediction Tasks: Set up a Spark environment for large-scale data processing. Implement scalable algorithms for sorting, searching, and filtering the dataset. Analyze the performance of your algorithms on different dataset sizes and optimize for scalability.	
--	--	--

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Assignment evaluation pattern:

- Correctness and Accuracy (30%) - Correct Solution and Implementation.
- Effectiveness and Efficiency (25%) - Algorithm Efficiency and Performance Metrics.
- Analytical Depth (25%) - Problem Understanding and Solution Analysis.
- Justification and Comparisons (20%) - Choice Justification and Comparative Analysis.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement data preprocessing and cleaning techniques to prepare raw data for analysis, ensuring the quality and reliability of the datasets.	K3
CO2	Perform exploratory data analysis (EDA) and create insightful visualizations that help in understanding the underlying patterns and trends in the data.	K4
CO3	Develop predictive models using various regression and classification algorithms, and optimize them for better performance, applying appropriate evaluation metrics.	K5
CO4	Implement scalable algorithms using distributed computing frameworks like Apache Spark to process large datasets efficiently.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							2
CO2	3	3	3	3								2
CO3	3	3	3		3							2
CO4	3	3	3		3							2
CO5	3	3	3		3							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Algorithms for Data Science Hardcover	Brian Steele, John Chandler, Swarna Reddy	Springer International	1/e, 2016
2	Mining of Massive Datasets	Jure Leskovec, Anand Rajaraman, Jeff Ullman	Cambridge University Press	2/e, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Foundations of Data Science	Avrim Blum, John Hopcroft and Ravi Kannan	Cambridge University Press	1/e, 2020
2	The Elements Of Statistical Learning: Data Mining, Inference, And Prediction	Trevor Hastie, Robert Tibshirani and Jerome Friedman	Springer	9/e, 2017
3	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber and Jian Pei Professor	Morgan Kaufmann	3/e, 2011
4	Data Mining and Predictive Analytics	Daniel T. Larose	Wiley	2/e, 2015
5	Hadoop for Dummies	Dirk Deroos, Paul C. Zikopoulos, Roman B. Melnyk, Bruce Brown, Rafael Coss	Wiley	1/e, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/104/106104189/ https://onlinecourses.nptel.ac.in/noc20_cs92/preview
2	https://archive.nptel.ac.in/courses/106/104/106104189/ https://onlinecourses.nptel.ac.in/noc20_cs92/preview
3	https://archive.nptel.ac.in/courses/106/104/106104189/ https://onlinecourses.nptel.ac.in/noc20_cs92/preview
4	https://archive.nptel.ac.in/courses/106/104/106104189/ https://nptel.ac.in/courses/106105186 https://archive.nptel.ac.in/courses/106/106/106106142/



DATA SCIENCE FOR ENGINEERS

PROF. RAGUNATHAN RENGASAMY

Department of Chemical Engineering
IIT Madras

PROF. SHANKAR NARASIMHAN

Department Of Chemical Engineering
IIT Madras

PRE-REQUISITES : 10 hrs of pre-course material will be provided, learners need to practise this to be ready to take the course.

INTENDED AUDIENCE : Any interested learner

INDUSTRY SUPPORT : HONEYWELL, ABB, FORD, GYAN DATA PVT. LTD

COURSE OUTLINE :

Learning Objectives :

- Introduce R as a programming language
- Introduce the mathematical foundations required for data science
- Introduce the first level data science algorithms
- Introduce a data analytics problem solving framework
- Introduce a practical capstone case study

Learning Outcomes:

- Describe a flow process for data science problems (Remembering)
- Classify data science problems into standard typology (Comprehension)
- Develop R codes for data science solutions (Application)
- Correlate results to the solution approach followed (Analysis)
- Assess the solution approach (Evaluation)
- Construct use cases to validate approach and identify modifications required (Creating)

ABOUT INSTRUCTOR :

Prior to joining IIT Madras as a professor, **Prof.Rengaswamy** was a professor of Chemical Engineering and Co-Director of the Process Control and Optimization Consortium at Texas Tech University, Lubbock, USA. He was also a professor and associate professor at Clarkson University, USA and an assistant professor at IIT Bombay. His major research interests are in the areas of fault detection and diagnosis and development of data science algorithms for manufacturing industries.

Prof.Shankar Narasimhan is currently a professor in the department of Chemical Engineering at IIT Madras. His major research interests are in the areas of data mining, process design and optimization, fault detection and diagnosis and fault tolerant control. He has co-authored several important papers and a book titled Data Reconciliation and Gross Error Detection: An Intelligent Use of Process Data which has received critical appreciation in India and abroad.

COURSE PLAN :

Week 1: Course philosophy and introduction to R

Week 2: Linear algebra for data science

1. Algebraic view - vectors, matrices, product of matrix & vector, rank, null space, solution of over-determined set of equations and pseudo-inverse)

2. Geometric view - vectors, distance, projections, eigenvalue decomposition

Week 3: Statistics (descriptive statistics, notion of probability, distributions, mean, variance, covariance, covariance matrix, understanding univariate and multivariate normal distributions, introduction to hypothesis testing, confidence interval for estimates)

Week 4: Optimization

Week 5: 1. Optimization

2. Typology of data science problems and a solution framework

Week 6: 1. Simple linear regression and verifying assumptions used in linear regression

2. Multivariate linear regression, model assessment, assessing importance of different variables, subset selection

Week 7: Classification using logistic regression

Week 8: Classification using kNN and k-means clustering

MOOC APPROVAL PROPOSAL

KTU Course Name	Parallel Algorithms
Course Category	Program Elective 5
Course Code	PECST759
MOOC Course Name	Parallel Computer Architecture
NPTEL Subject ID	106103359
Course ID	noc26-cs25
Course URL	https://nptel.ac.in/courses/106103359
COORDINATORS	Prof. Hemangee K.; Kapoor
Offering Institute	IIT Guwahati
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

PARALLEL ALGORITHMS

(Common to CS/CM/CD/AM)

Course Code	PECST759	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCST303 PCCST502	Course Type	Theory

Course Objectives:

1. To develop a comprehensive understanding of parallel computing principles and architectures by studying various types of parallelism, such as data and task parallelism, and analyzing different computing architectures.
2. To implement and evaluate parallel algorithms for fundamental operations, such as matrix addition and multiplication, using performance metrics like speedup and scalability, while gaining hands-on experience with parallel programming models and tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Parallel Computing - Overview of parallel computing and its importance, Types of parallelism: data parallelism, task parallelism, Parallel computing architectures: SIMD, MIMD, shared memory, distributed memory. Parallel Programming Models - Parallel programming models: Parallel Random Access Machine (PRAM), bulk synchronous parallel (BSP), LogP, Shared memory vs. distributed memory models; Performance Metrics - Performance metrics for parallel algorithms: speedup, efficiency, scalability, Amdahl's Law and Gustafson's Law.	9
2	Parallel Algorithms for Basic Operations - Parallel algorithms for matrix addition, matrix multiplication, and reduction, Parallel prefix sum (Parallel scan) algorithms. Case Studies of Parallel Addition, Multiplication, Reduction, and Prefix Sum in Modern Computing Systems; Parallel Sorting Algorithms - Parallel sorting algorithms: parallel merge sort, parallel quicksort, bitonic merge sort, Comparison of parallel sorting techniques.	9
3	Parallel Graph Algorithms - Parallel algorithms for graph traversal: BFS, DFS, Parallel algorithms for minimum spanning tree (MST) and shortest path.	9

	Parallel Search Algorithms - Parallel search algorithms: parallel binary search, parallel search trees, Applications and analysis.	
4	Parallel Programming with OpenMP - Introduction to OpenMP, Parallel programming constructs in OpenMP, Performance tuning and optimization Parallel Programming with MPI - Introduction to MPI, Message passing model and MPI basics, Advanced MPI features and applications Parallel Numerical Algorithms - Solving linear systems: parallel Gaussian elimination, parallel LU decomposition, Parallel algorithms for eigenvalue problems, Applications and analysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and articulate the fundamental principles and architectures of parallel computing.	K2
CO2	Implement and evaluate parallel algorithms for basic operations such as sorting and searching.	K3
CO3	Develop and analyze parallel algorithms for complex problems, including graph and numerical algorithms.	K3
CO4	Apply parallel programming techniques to real-world problems and assess the efficiency and performance of parallel solutions.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2								3
CO2	3	3	3	2								3
CO3	3	3	3	3								3
CO4	3	3	3	3			2	2				3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Parallel Computing	Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar	Addison-Wesley	2/e, 2003
2	Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers	Barry Wilkinson and Michael Allen	Pearson India	2/e, 2006
3	An Introduction to Parallel Algorithms	Joseph Jaja	Addison-Wesley Professional	1/e, 1992
4	Parallel Algorithms	Henri Casanova, Arnaud Legrand, Yves Robert	Chapman and Hall/CRC	1/e, 2020
5	Parallel Scientific Computing in C++ and MPI	George Em Karniadakis and Robert M. Kirby II	Cambridge University Press	1/e, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Parallel Programming for Multicore and Cluster Systems	Thomas Rauber, Gudula Rünger	Springer	3/e, 2023
2	Using OpenMP: Portable Shared Memory Parallel Programming	Barbara Chapman, Gabriele Jost, Ruud van der Pas	MIT Press	1/e, 2007
3	Using MPI: Portable Parallel Programming with the Message-Passing Interface	William Gropp, Ewing Lusk, Anthony Skjellum	MIT Press	3/e, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106112/
2	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120
3	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120
4	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120



PARALLEL COMPUTER ARCHITECTURE

PROF. HEMANGEE K. KAPOOR

Department of Computer Science and Engineering
IIT Guwahati

PRE-REQUISITES : Final year undergraduate as well as post-graduate students in the disciplines of CSE, IT, CS, ECE, EEE, etc. Can benefit from this course. A basic knowledge in Computer Organisation is sufficient to take this subject.

INTENDED AUDIENCE : Final year UG as well as PG

INDUSTRY SUPPORT : Intel, Micron, AMD, IBM, NVIDIA, HP, Samsung, etc.

COURSE OUTLINE :

With the emergence of application domains requiring large data processing as well as faster compute performance, parallel computing has become a critical component. This involves the use of multi-core processors as well as tightly integrated clusters of PCs and workstations. The most exciting development is the convergence of traditionally disparate approaches of shared memory, message passing, and data driven computing onto the same platform.

As the parallel applications communicate with each other using shared data; it is important to correctly manage the shared data within the memory hierarchy to avoid inconsistency. Here, we need to understand the mechanisms used for data access, communication and coordination of work and its correct implementation.

This course will introduce the field of parallel architectures and discuss in-depth shared memory management for parallel architectures. We will also learn about the interconnection topologies and routing methods which are important for connection and communication of the multiple cores executing the parallel applications. The course will conclude with an overview of various parallel architectures and on-going research directions.

ABOUT INSTRUCTOR :

Prof. Hemangee K. Kapoor is a Professor in the Department of Computer Science Engineering at the Indian Institute of Technology Guwahati, Assam. She has over 18 years of teaching experience at both undergraduate and postgraduate levels. She received her B.Eng. degree in Computer Engineering from the College of Engineering, Pune, the M.Tech. degree in CSE from the IIT Bombay, and the Ph.D. degree in CSE from London South Bank University, U.K. Prof. Kapoor has been teaching subjects related to computer organisation, digital design, advanced computer architecture, parallel computer architecture and hardware lab. She has also given expert lectures in FDPs and TEQIP events. Her research spans in the broad area of computer architecture and specially in multicore caches, emerging non-volatile memory technologies, network-on-chip and design of accelerators for neural networks. Till date she has graduated 6 PhD and 39 MTech students. She is a Senior Member of the IEEE and ACM. She has published her work in reputed journals and conferences. She is an Associate Editor for the Journal of Systems Architecture and IEEE Design and Test. Prof. Kapoor has served in the ACM India council: as a member (2014-22) and as Vice President (2020-22). She serves as a member on the ACM international council for Diversity, Equity and Inclusion (ACM-DEI) (2021-23). More details can be found on website: URL: <https://iitg.ac.in/hemangee/>

COURSE PLAN :

Week 1 : Introduction to Parallel Architectures

Week 2 : Parallel Programming models and Architectures

Week 3 : Memory Hierarchy-Cache and Virtual memory

Week 4 : Overview of Cache coherence, Coherence Protocols- Snooping, Directory based protocols, VI protocol

Week 5 : MSI, MESI, Dragon protocol and Correctness of coherence protocols- Types of cache misses, update vs invalidate protocol

Week 6 : Snoop based multiprocessor design, Single and multi-level cache with atomic bus

Week 7 : Snoop based multiprocessor design, Single and multi-level cache with split transaction bus

Week 8 : Scalable shared memory systems: Directory coherence protocols- Memory based, cache based, correctness

Week 9 : Case study: Origin- Architecture, protocol, correctness; Sequent NUMA Q- Architecture, protocol, correctness

Week 10 : Memory consistency models- Sequential, Relaxed consistency models

Week 11 : Synchronization- LL-SC, point to point, barrier synchronization

Week 12 : Interconnects- Introduction, Topologies, routing, flow control

MOOC APPROVAL PROPOSAL

KTU Course Name	Fuzzy Systems
Course Category	Program Elective 5
Course Code	PECST753
MOOC Course Name	FUZZY SETS, LOGIC AND SYSTEMS & APPLICATIONS
NPTEL Subject ID	108104157
Course ID	noc26-ee38
Course URL	https://nptel.ac.in/courses/108104157
COORDINATORS	Prof. Nishchal Kumar; Verma
Offering Institute	IIT Kanpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

FUZZY SYSTEMS

(Common to CS/CA)

Course Code	PECST753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the concepts of fuzziness and its use in building better solutions to problems.
2. To understand the basic concepts of fuzzy sets, fuzzy relations, fuzzy logic and building of fuzzy approximation-based solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic Fuzzy Set Theory :- Introduction - Uncertainty, Imprecision and Vagueness. Crisp vs Fuzzy sets. Representation of Fuzzy sets. Membership Functions – Types, Basic operations - dilation, concentration, normalization, Linguistic hedges. Properties of fuzzy set - Level Sets - Alpha cut representation. Operations on fuzzy sets- fuzzy complement, fuzzy intersection, fuzzy union, aggregation operations	9
2	Fuzzy Relations :- Operations on Fuzzy relations: union, intersection, complement, cartesian product. Fuzzy composition- Max- min, Max – product. Extension Principle- Fuzzy arithmetic – fuzzy numbers, arithmetic operations on fuzzy numbers. Fuzzy Reasoning – Generalized Modus Ponens (GMP) and Generalized Modus Tollens (GMT).	9

3	Fuzzification and Defuzzification Methods :- Fuzzy inference – Zadeh rule, Mamdani rule. Development of membership Functions – Intuition, Inference, Rank ordering, Inductive reasoning. Defuzzification to Scalars - Max membership principle, Centroid method, Weighted average method, Mean max membership, Center of sums, Center of largest area, First (or last) of maxima.	9
4	Fuzzy Inference Systems :- Approximate Reasoning, Fuzzy (Rule-Based) Systems – Multiple conjunctive antecedents, Multiple disjunctive antecedents, Aggregation of fuzzy rules, Graphical Techniques of Inference. Fuzzy Controllers -Mamdani FIS, Larsen Model.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)
Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fuzzy logic based problem solving	K2
CO2	Summarize the concepts of crisp sets, crisp relations, crisp logic with fuzzy sets, fuzzy relations and fuzzy logic	K3
CO3	Develop fuzzy systems by selecting appropriate membership functions, fuzzification and defuzzification methods	K3
CO4	Develop solutions using graphical and rule-based methods	K3
CO5	Make use of fuzzy logic inference to solve real world problems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									2
CO2	3	1	1									2
CO3	3	3	2	1								2
CO4	3	3	2	1								2
CO5	3	3	2	2	1							2

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuzzy Logic with Engineering Applications	Timothy J. Ross	John Wiley and Sons	3/e, 2010
2	Fuzzy Sets and Fuzzy Logic: Theory and Applications	George J. Klir and Bo Yuan	Pearson	1/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Fuzzy Sets, Fuzzy Logic, and Fuzzy Control Systems	Guanrong Chen, Trung Tat Pham	CRC Press	1/e, 2019
2	Discrete Mathematics and Its Applications with Combinatorics and Graph Theory	Kenneth H. Rosen	MGH	7/e, 2011
3	Discrete Mathematical Structures with Applications to Computer Science	Trembly J.P, Manohar R	TataMc Graw Hill	1/e, 2003
4	Discrete Mathematical Structures	Bernard Kolman, Robert C. Busby, Sharan Cutler Ross,	Pearson	1/e, 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108104157



FUZZY SETS, LOGIC AND SYSTEMS & APPLICATIONS

PROF. NISHCHAL KUMAR VERMA

Department of Electrical Engineering
IIT Kanpur

INTENDED AUDIENCE : UG, PG Students, industry professionals, researchers etc.

COURSE OUTLINE :

The course is designed to give a solid grounding of fundamental concepts of fuzzy logic and its applications. The level of the course is chosen to be such that all students aspiring to be a part of computational intelligence directly or indirectly in near future should get these concepts.

ABOUT INSTRUCTOR :

Dr. Nishchal Kumar Verma is a Professor in the Department of Electrical Engineering at the Indian Institute of Technology Kanpur. He has received several awards, including the Smt. Lata and K.G. Karandikar Faculty Chair Professor (2024), the Teaching Excellence (2024), the prestigious research fellowship from UNEC, Azerbaijan (2023), a research fellowship from University of Tennessee, Memphis, USA (2022, 2010), and the Devendra Shukla Young Faculty Research Fellowship (2013). Dr. Verma specializes in the theoretical developments of Artificial Intelligence, Machine Learning, and Deep Learning, applying these concepts to Electrical Engineering and various interdisciplinary fields. His research areas include Large Language Models, Intelligent Instrumentation, Control, Automation, Computer Vision, Prognosis and Health Management of Machines, Bioinformatics, and Cyber-Physical Systems, among others. Dr. Verma and his team have developed several AI-based technologies for the BOEING Company, USA, and organized numerous national and international workshops, conferences, seminars, brainstorming sessions, and short-term courses. He has published over 270 research papers in reputable peer-reviewed journals and conferences and has authored, co-authored, or edited six books. He has successfully completed a significant number of research projects funded by various agencies, including The BOEING Company, DST, DRDO, MHRD, MeitY, CSIR, IIT Kanpur, MCIT, SFTIG, and VTOL etc. With over 20 years of experience in the fields of Artificial Intelligence and Machine Learning, Dr. Verma serves as an Associate Editor for key publications, including IEEE Transactions on Artificial Intelligence (2020-2024), IEEE Transactions on Neural Networks and Learning Systems (2019-2024), and IEEE Computational Intelligence Magazine (2015-2020).

COURSE PLAN :

Week 1: Introduction and Fuzzy Sets Theory

Week 2: Membership Functions

Week 3: Set Theoretic Operations

Week 4: Fuzzy Arithmetic

Week 5: Fuzzy Relations

Week 6: Fuzzy Inference Systems I

Week 7: Fuzzy Inference Systems II

Week 8: Wang and Mendel Model

Week 9: TSK Model

Week 10: Fuzzifiers and Defuzzifiers

Week 11: ANFIS Architecture

Week 12: Fuzzy Systems and Machine Learning

MOOC APPROVAL PROPOSAL

KTU Course Name	High Performance Computing
Course Category	Program Elective 5
Course Code	PECST757
MOOC Course Name	High Performance Scientific Computing
NPTEL Subject ID	111101611
Course ID	noc26-ma16
Course URL	https://nptel.ac.in/courses/111101611
COORDINATORS	Prof. Shivasubramanian Gopalakrishnan;; Prof. Om Jadhav;; Prof. S Vamshi Krishna;; Prof. Ashish Kuvelkar
Offering Institute	IIT Bombay
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

HIGH PERFORMANCE COMPUTING

(Common to CS/CR/CM/CD/CA/AM/AD)

Course Code	PECST757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To Gain an understanding of the modern processor architectures.
2. To Give an introduction to parallel programming using OpenMP and MPI.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modern processors: Stored-program computer architecture- <i>General-purpose cache-based microprocessor architecture</i> - Performance metrics and benchmarks -Moore's Law - Pipelining - Super scalarity - SIMD - <i>Memory hierarchies</i> - Cache , Cache mapping, Prefetch, Multicore processors - Multithreaded processors - <i>Vector processors</i> - Design principles - Maximum performance estimates - Programming for vector architectures.	9
2	Parallel computers - Taxonomy of parallel computing paradigms - <i>Shared-memory computers</i> - Cache coherence - UMA, ccNUMA, Distributed-memory computers - Hierarchical (hybrid) systems - <i>Networks</i> - Basic performance characteristics of networks, Buses, Switched and fat-tree networks - Mesh networks - Hybrids.	9
3	Shared-memory parallel programming with OpenMP:- <i>Short introduction to OpenMP</i> - Parallel execution - Data scoping - OpenMP worksharing for loops - Synchronization, Reductions, Loop scheduling, Tasking,Miscellaneous, Case study: OpenMP-parallel Jacobi algorithm	9

4	Distributed-memory parallel programming with MPI:- Message passing - <i>A short introduction to MPI</i> , A simple example, Messages and point-to-point communication, Collective communication, Nonblocking point-to-point communication, Virtual topologies. <i>Example-MPI parallelization of a Jacobi solver</i> - MPI implementation - Performance properties.	9
----------	---	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe parallel computing architectures supported by modern processors.	K2
CO2	Classify parallel computing paradigms and network topologies.	K2
CO3	Implement shared-memory parallel programming with OpenMP.	K3
CO4	Design and implement parallel algorithms using distributed-memory parallel programming with MPI	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	3	3	2								3
CO4	3	3	3	2								3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to High Performance Computing for Scientists and Engineers	Georg Hager Gerhard Wellein	CRC Press	1/e, 2011
2	High Performance Computing: Modern Systems and Practices	Thomas Sterling, Maciej Brodowicz, Matthew Anderson	Morgan Kaufmann	1/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Parallel and High-Performance Computing	Robert Robey Yuliana Zamora	Manning Publications	1/e, 2021
2	High-Performance Computing	Charles Severance Kevin Dowd	O'Reilly Media	2/e, 1998
3	Computer Architecture And Parallel Processing	Kai Hwang Faye Alaye Briggs	McGraw-Hill	1/e, 1984
4	Computer Architecture: A Quantitative Approach	John L. Hennessy David A. Patterson	Morgan Kaufman	6/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106108055
2	https://nptel.ac.in/courses/106108055
3	https://nptel.ac.in/courses/106108055
4	https://nptel.ac.in/courses/128106014

HIGH PERFORMANCE SCIENTIFIC COMPUTING

MULTIFACULTY

PRE-REQUISITES : Basic course on programming and applied mathematics

INTENDED AUDIENCE : Researchers, graduate students , postdocs working in the area of computational science.

INDUSTRY SUPPORT : The industries in which this would be useful for is aerospace, automotive, defence, chemical , electrical, materials, biomedical and nuclear industries which employ simulation technologies.

COURSE OUTLINE :

Scientific computing has become an important third axis in the development of science in conjunction with theoretical and experimental studies. The tremendous growth in computing power, especially with high performance computing (HPC) clusters, over the last few decades have opened up opportunities to computationally study phenomena which were earlier beyond reach. Fluid Dynamics, Electromagnetics, Astrophysics, Biology, Finance etc are few of the domains which are greatly aided by scientific computing. There is a dire need to train researchers and graduate students in the effective use and programming practices of these HPC clusters. This course will aim to fill the gap in understanding and use of such systems.

ABOUT INSTRUCTOR :

Prof. Shiva Gopalakrishnan in 2012 joined the Indian Institute of Technology Bombay as a faculty member and currently is an Associate Professor in the Department of Mechanical Engineering at the Indian Institute of Technology Bombay. He obtained his Ph.D from the University of Massachusetts - Amherst and subsequently was a National Research Council Postdoctoral Fellow in the Department of Applied Mathematics at the Naval Postgraduate School, Monterey, California. Shiva Gopalakrishnan has developed a graduate level course on High Performance Scientific Computing. Scientific computing has become an important third axis in the development of science in conjunction with theoretical and experimental studies. The tremendous growth in computing power, especially with high performance computing (HPC) clusters, over the last few decades have opened up opportunities to computationally study phenomena which were earlier beyond reach. Fluid Dynamics, Electromagnetics, Astrophysics, Biology, Finance etc are few of the domains which are greatly aided by scientific computing. There is a dire need to train researchers and graduate students in the effective use and programming practices of these HPC clusters. This course fills the gap in understanding and use of such systems. The curriculum introduces the basics of modern computer architecture, parallel computing machines and parallel programming. It covers an in depth discussion on different parallel programming paradigms and associated techniques for efficient programming and model development. The success of this course is shown in the enrolment which is in excess of 120 students every time it is offered at the Indian Institute of Technology Bombay. In addition this course has been offered in adapted forms at the University of Florida Gainesville and Koc University, Istanbul, Turkey where it was well received.

Prof. Om Jadhav has received his master's degree in Telecom Technology from IIT Delhi. He is working as a Scientist D, at Centre for Development of Advanced Computing, where he is associated with HPC-Technologies team at C-DAC Pune. His areas of expertise include, HPC Application's Optimization and management on HPC clusters along with understanding of parallel programming, distributed computing, and HPC architectures. He is currently working on different projects associated with Research and Development activities in HPC and AI domains.

Prof. Ashish Kuvelkar obtained Masters degree in Electrical Engineering from V. J. T. I., Bombay University. He is working as a Scientist G, at Centre for Development of Advanced Computing, where he is associated with HPC-Technologies team at C-DAC Pune. Previously, he worked for Hardware Division of Patni Computers for 9 years. At C-DAC, he has been associated with design of hardware subsystems for various generation of PARAM supercomputers. He also contributes to training activities of Advanced Computing Training School of C-DAC, which conducts Post-Graduate Diploma courses in various specializations. Currently, he is involved in the HPC aware Human Resource development activities of the National Supercomputing Mission.

Prof. Ashish Kuvelkar obtained Masters degree in Electrical Engineering from V. J. T. I., Bombay University. He is working as a Scientist G, at Centre for Development of Advanced Computing, where he is associated with HPC-Technologies team at C-DAC Pune. Previously, he worked for Hardware Division of Patni Computers for 9 years. At C-DAC, he has been associated with design of hardware subsystems for various generation of PARAM supercomputers. He also contributes to training activities of Advanced Computing Training School of C-DAC, which conducts Post-Graduate Diploma courses in various specializations. Currently, he is involved in the HPC aware Human Resource development activities of the National Supercomputing Mission.

COURSE PLAN:

Week 1: Introduction to high performance computing and scientific computing. The need for HPSC.

Week 2: Processor performance. Memory hierarchy. Multi-core processing and Vector computing

Week 3: Introduction to parallel programming concepts and parallel algorithms

Week 4: Effective use of command line Linux. Bash scripting

Week 5: Use of version control systems such as Git/SVN/Mercurial.

Week 6: Introduction to OpenMP and thread programming

Week 7: Introduction to MPI programming

Week 8: Introduction to GPGPU / Vector

Week 9: Effective use of debuggers and parallel

Week 10: Performance analysis of parallel

Week 11: Use of toolkits such as BLAS, LAPACK,

Week 12: Advanced scientific visualization.

MOOC APPROVAL PROPOSAL

KTU Course Name	Parallel Algorithms
Course Category	Program Elective 5
Course Code	PECST759
MOOC Course Name	Parallel Algorithms
NPTEL Subject ID	106103188
Course URL	https://nptel.ac.in/courses/106103188
COORDINATORS	Prof. Sajith Gopalan
Offering Institute	IIT Guwahati
Duration	12 weeks
Platform	-

SEMESTER S7

PARALLEL ALGORITHMS

(Common to CS/CM/CD/AM)

Course Code	PECST759	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCST303 PCCST502	Course Type	Theory

Course Objectives:

1. To develop a comprehensive understanding of parallel computing principles and architectures by studying various types of parallelism, such as data and task parallelism, and analyzing different computing architectures.
2. To implement and evaluate parallel algorithms for fundamental operations, such as matrix addition and multiplication, using performance metrics like speedup and scalability, while gaining hands-on experience with parallel programming models and tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Parallel Computing - Overview of parallel computing and its importance, Types of parallelism: data parallelism, task parallelism, Parallel computing architectures: SIMD, MIMD, shared memory, distributed memory. Parallel Programming Models - Parallel programming models: Parallel Random Access Machine (PRAM), bulk synchronous parallel (BSP), LogP, Shared memory vs. distributed memory models; Performance Metrics - Performance metrics for parallel algorithms: speedup, efficiency, scalability, Amdahl's Law and Gustafson's Law.	9
2	Parallel Algorithms for Basic Operations - Parallel algorithms for matrix addition, matrix multiplication, and reduction, Parallel prefix sum (Parallel scan) algorithms. Case Studies of Parallel Addition, Multiplication, Reduction, and Prefix Sum in Modern Computing Systems; Parallel Sorting Algorithms - Parallel sorting algorithms: parallel merge sort, parallel quicksort, bitonic merge sort, Comparison of parallel sorting techniques.	9
3	Parallel Graph Algorithms - Parallel algorithms for graph traversal: BFS, DFS, Parallel algorithms for minimum spanning tree (MST) and shortest path.	9

	Parallel Search Algorithms - Parallel search algorithms: parallel binary search, parallel search trees, Applications and analysis.	
4	Parallel Programming with OpenMP - Introduction to OpenMP, Parallel programming constructs in OpenMP, Performance tuning and optimization Parallel Programming with MPI - Introduction to MPI, Message passing model and MPI basics, Advanced MPI features and applications Parallel Numerical Algorithms - Solving linear systems: parallel Gaussian elimination, parallel LU decomposition, Parallel algorithms for eigenvalue problems, Applications and analysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and articulate the fundamental principles and architectures of parallel computing.	K2
CO2	Implement and evaluate parallel algorithms for basic operations such as sorting and searching.	K3
CO3	Develop and analyze parallel algorithms for complex problems, including graph and numerical algorithms.	K3
CO4	Apply parallel programming techniques to real-world problems and assess the efficiency and performance of parallel solutions.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2								3
CO2	3	3	3	2								3
CO3	3	3	3	3								3
CO4	3	3	3	3			2	2				3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Parallel Computing	Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar	Addison-Wesley	2/e, 2003
2	Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers	Barry Wilkinson and Michael Allen	Pearson India	2/e, 2006
3	An Introduction to Parallel Algorithms	Joseph Jaja	Addison-Wesley Professional	1/e, 1992
4	Parallel Algorithms	Henri Casanova, Arnaud Legrand, Yves Robert	Chapman and Hall/CRC	1/e, 2020
5	Parallel Scientific Computing in C++ and MPI	George Em Karniadakis and Robert M. Kirby II	Cambridge University Press	1/e, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Parallel Programming for Multicore and Cluster Systems	Thomas Rauber, Gudula Rünger	Springer	3/e, 2023
2	Using OpenMP: Portable Shared Memory Parallel Programming	Barbara Chapman, Gabriele Jost, Ruud van der Pas	MIT Press	1/e, 2007
3	Using MPI: Portable Parallel Programming with the Message-Passing Interface	William Gropp, Ewing Lusk, Anthony Skjellum	MIT Press	3/e, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106112/
2	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120
3	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120
4	https://archive.nptel.ac.in/courses/106/106/106106112/ https://nptel.ac.in/courses/106104120



COMPUTER SCIENCE & ENGINEERING

PARALLEL ALGORITHMS



PROF. SAJITH GOPALAN

Dept. of Computer Science and Engineering
IIT Guwahati

TYPE OF COURSE : New | Elective | UG/PG

COURSE DURATION : 12 weeks (28 Jan'19 - 19 Apr'19)

INTENDED AUDIENCE : B.E/B.Tech, M.E/M.Tech

EXAM DATE : 28 April 2019

PRE-REQUISITES : Courses in Data Structures, Algorithms and Discrete Mathematics

COURSE OUTLINE :

A conventional algorithm uses a single processing element. A parallel algorithm assumes that there are multiple processors. These processors may communicate with each other using a shared memory or an interconnection network. An algorithm designed for a large number (for example, a polynomial in the problem size) of processors can be simulated on a machine with a small number of processor for a trade off on time, and therefore is of practical value, while at the same time allowing us to test the limits of parallelism. Many algorithmic design techniques in the parallel setting will be explored. Parallel complexity theory will also be briefly studied.

ABOUT INSTRUCTOR :

Professor Sajith Gopalan has been in the faculty of Computer Science and Engineering, IIT Guwahati since 1997. PhD (IIT Kanpur, 1998), MTech (IIT Kanpur, 1993), BTech (REC Calicut, 1991)

COURSE PLAN :

Week 01 : Theoretical models of parallel computation: PRAM, interconnection networks

Week 02 : Performance of parallel algorithms, Basic techniques

Week 03 : Basic techniques (cont'd)

Week 04 : Comparator Networks. Odd Even Merge Sort. Bitonic Sort Merge Sort.

Week 05 : Optimal List ranking, applications

Week 06 : Algorithms for searching, merging and sorting. Cole's Merge Sort

Week 07 : Cole's Merge Sort(cont'd), Graph algorithms

Week 08 : Graph Algorithms (cont'd), Linear Array, Meshes

Week 09 : Sorting in meshes, Hypercube algorithms, Butterfly network, CCC, Benes network

Week 10 : Butterfly network, ccc, benes network

Week 11 : Limits to parallelizability. Lower bounds

Week 12 : Limits to parallelizability. NC-reductions, P-completeness.

MOOC APPROVAL PROPOSAL

KTU Course Name	Optimization Techniques
Course Category	Open Elective 2
Course Code	OEECT723
MOOC Course Name	Optimization from fundamentals
NPTEL Subject ID	112101298
Course ID	noc26-me09
Course URL	https://nptel.ac.in/courses/112101298
COORDINATORS	Prof. Ankur A. Kulkarni
Offering Institute	IIT Bombay
Duration	12 weeks
Platform	NPTEL

SEMESTER S7

OPTIMIZATION TECHNIQUES

Course Code	OEECT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Enable the learner to formulate engineering minima/maxima problems as optimization problems
2. Enable the learner to deploy various constrained and unconstrained optimization algorithms to obtain the minima/maxima of engineering problems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Engineering application of Optimization – Statement of an Optimization problem–Classification, Review of basic calculus concepts –Stationary points; Functions of single and two variables; Convexity and concavity of functions –Definition of Global and Local optima – Optimality criteria, Linear programming methods for optimum design – Standard form of linear programming (LP) problem; Canonical form of LP problem; Simplex Method, Duality, Application of LPP models in engineering	9
2	Optimization algorithms for solving unconstrained nonlinear optimization problems – Search based techniques: Direct search: Fibonacci and golden section search , Hookes and Jeeves , Gradient based method: Newton's method	9

3	Optimization algorithms for solving constrained optimization problems– direct methods – penalty function methods, barrier method -Optimization of function of multiple variables subject to equality constraints; Lagrangian function– Inequality constrained techniques-KKT conditions-constrained steepest descent method	9
4	Modern methods of Optimization– Metaheuristic techniques: Genetic Algorithms – Simulated Annealing – Particle Swarm optimization –Ant colony optimization– : Use of Matlab/Scilab to solve optimization problem	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Formulate an optimization problem to optimize an engineering application using the principles of basic calculus.	K2
CO2	Apply the Simplex method to solve a linear programming problem	K3
CO3	Solve the unconstrained optimization problems using gradient based method.	K3
CO4	Apply the various optimization techniques to solve a constrained optimization problem	K3
CO5	Use metaheuristic algorithms to solve constrained and unconstrained	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	3	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	3									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Optimization, Theory and Practice	S.S RAO	New Age International Publishers	4 th Edition ,2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Optimization Techniques and Applications with Examples	Xin-She Yang	John Wiley & Sons	2018
2	Optimization for Engineering Design Algorithms and Examples	Deb K	Prentice Hall India	2000
3	Introduction to Optimization Design	Arora J	Elsevier Academic Press, New Delhi	2004
4	Linear Programming	Hardley G	Narosa Book Distributors Private Ltd	2002
5	Genetic Algorithms and engineering optimization	Mitsuo Gen, Runwei Cheng	John Wiley & Sons	2002
6	An introduction to optimization	Edwin KP Chong, Stanislaw, H Hak	John Wiley & Sons	Fourth Edition, 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL https://www.youtube.com/watch?v=a2QgdDk4Xjw
2	NPTEL https://www.youtube.com/watch?v=dPQKltPBLfc
3	NPTEL https://www.youtube.com/watch?v=qY-gKL7GxYk
4	NPTEL https://www.youtube.com/watch?v=Z_8MpZeMdD4 https://www.youtube.com/watch?v=FKBgCpJlX48



OPTIMIZATION FROM FUNDAMENTALS

PROF. ANKUR A. KULKARNI

Department of Systems and Control Engineering
IIT Bombay

INTENDED AUDIENCE : Mathematics, any engineering and science discipline

COURSE OUTLINE :

This course will cover the topic of Optimization from its fundamentals. It will start with an overview of real analysis and convexity. With this base it will cover linear programming, convex optimization and nonlinear programming, complementarity problems and algorithms for the same. We will end with dynamic optimization.

ABOUT INSTRUCTOR :

Prof. Ankur A. Kulkarni is an Associate Professor and the Kelkar Family Chair in Quantitative Finance at the Indian Institute of Technology Bombay (IITB). He is a systems theorist with an interest in decision making in distributed, decentralised and strategic environments, with informational complexities, which he investigates using the lens of game theory, information theory, control theory, machine learning, and mathematical optimization. His current focus is on strategic inference, stealth, privacy, information elicitation and nudging. He has published 30 papers in international journals and more than 30 papers in international conferences. He has been a consultant to the Securities and Exchange Board of India wherein he was solely responsible for suggesting regulatory interventions for high frequency algorithmic trading. He presently serves on the IT-Project Advisory Board of SEBI and is responsible for advising SEBI on utilizing advanced technologies such as AI/ML and data analytics and guiding data-related policies for internal use and public use. He is also an advisor to the Tata Consultancy Services and on the technical advisory committee of Maha-IT, a Govt of Maharashtra enterprise. He was previously a consultant to HDFC Life Insurance Company wherein he tackled the problem of design of incentives for sales agents; to Kotak Mahindra Bank Limited on anti-money laundering and anomaly detection, and to Bank of Baroda on smart cash management. He received his B.Tech. from IITB in 2006, followed by M.S. in 2008 and Ph.D. in 2010, both from the University of Illinois at Urbana-Champaign (UIUC). He was an Associate (from 2015--2018) of the Indian Academy of Sciences, Bangalore (a honour reserved for only 100 scientists under the age of 35 across all fields), he has been an editor for several conferences, a recipient of the INSPIRE Faculty Award of the Department of Science and Technology, Government of India, 2013, and of several Best Paper awards and the Excellence in Teaching Award at IIT Bombay. He has been a visitor to MIT in the USA, University of Cambridge in UK, NUS in Singapore, University of Paris, IISc in Bangalore and KTH in Sweden.

COURSE PLAN :

- Week 1:** Introduction to optimization and overview of real analysis
- Week 2:** Optimization over open sets
- Week 3:** Optimization over surface
- Week 4:** Transformation of optimization problems and convex analysis
- Week 5:** Introduction to linear programming
- Week 6:** Linear programming and duality
- Week 7:** Linear programming and duality
- Week 8:** Nonlinear and convex optimization
- Week 9:** Nonlinear and convex optimization
- Week 10:** Algorithms
- Week 11:** Algorithms
- Week 12:** Dynamic optimization

MOOC APPROVAL PROPOSAL

KTU Course Name	Robotics
Course Category	Open Elective 2
Course Code	OEMET722
MOOC Course Name	Robotics and Control : Theory and Practice
NPTEL Subject ID	112107289
Course ID	noc26-me72
Course URL	https://nptel.ac.in/courses/112107289
COORDINATORS	Prof. M. Felix Orlando; Prof. N. Sukavanam
Offering Institute	IIT Roorkee
Duration	8 weeks
Platform	NPTEL

SEMESTER S7

ROBOTICS

Course Code	OEMET722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course helps the student to the basic idea of Robots. Students are introduced to the basic design considerations of robots.
2. Concepts like trajectory planning and obstacle avoidance and kinematics of robots are introduced.
3. Discussion on various mobile robots and robotic manipulators are also included as part of the course to get an overall idea on robotics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots. Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers.	9
2	Direct Kinematics- Rotations-Fundamental and composite Rotations, Homogeneous coordinates, Translations and rotations, Composite homogeneous transformations, Screw transformations, Kinematic parameters, The Denavit-Hartenberg (D-H) representation, The arm equation, direct kinematics problems (up to 3DOF). Inverse kinematics- general properties of solutions and problems (up to	10

	3DOF).	
3	Manipulator Dynamics: Lagrange's formulation – Kinetic Energy expression, velocity Jacobian and Potential Energy expression, Generalised force, Euler-Lagrange equation, Dynamic model of planar and spatial serial robots up to 2 DOF, modelling including motor and gearbox.	8
4	Trajectory Planning. Joint space trajectory planning- cubic polynomial, linear trajectory with parabolic blends, trajectory planning with via points; Cartesian space planning, point-to-point vs. continuous path planning. Obstacle avoidance methods- Artificial Potential field, A* algorithms. Robot Control: The control problem, Single axis PID control-its disadvantages, PD gravity control, computed torque control	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise with anatomy, specifications and types of Robots	K2
CO2	Obtain forward and inverse kinematic models of robotic manipulators	K3
CO3	Plan trajectories in joint space & Cartesian space and avoid obstacles while robots are in motion	K4, K5
CO4	Develop a dynamic model and design the controller for robotic manipulators	K4, K6
CO5	Choose the appropriate Robotic configuration and list the technical specifications for robots used in different application	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	3
CO2	2	1	-	-	-	-	-	-	-	-	-	3
CO3	2	1	-	-	-	-	-	-	-	-	-	3
CO4	3	2	2	-	-	-	-	-	-	-	-	3
CO5	3	2	2	-	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Robotics – Analysis and Control	Robert. J. Schilling	Prentice Hall of India	1996
2	Introduction to Robotics (Mechanics and Control)	John. J. Craig	Pearson Education Asia	2002
3	Introduction to Robotics	S K Saha,	McGraw Hill Education	
4	Robotics and Control	R K Mittal	Tata McGraw Hill, New Delhi	2003
5	Robotics-Fundamental concepts and analysis	AshitavaGhosal	Oxford University Press	
6	Robotics Technology and Flexible Automation,	S. R. Deb		Second Edition,
7	Introduction to Autonomous Mobile Robots	Siegwart, Roland,	Cambridge, Mass.: MIT Press,	Second Edition,

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Robotics	Siciliano, Khatib	Springer	
2	Introduction to Robotics – Mechanics and Control	John J. Craig		
3	Modern Robotics Mechanics, Planning and Control	. Kevin M. Lynch, Frank C. Park,		
4	Robotics Modelling, Planning and Control	Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, Giuseppe Oriolo	Springer	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/112/105/112105249/
3	https://archive.nptel.ac.in/courses/112/105/112105249/
4	https://archive.nptel.ac.in/courses/112/105/112105249/



ROBOTICS AND CONTROL: THEORY AND PRACTICE

PROF. N. SUKAVANAM

Department of Mathematics
IIT Roorkee

PROF .M. FELIX ORLANDO

Department of Electrical Engineering
IIT Roorkee

PRE-REQUISITES : Basic Mathematics

INTENDED AUDIENCE : Electrical Engineering, Computer Science Engineering, Mechanical Engineering, Electronics and Communication Engineering, Mathematics students

COURSE OUTLINE :

Robotics has stimulated an growing interest among a wide range of scholars, researchers and students due to its interdisciplinary characteristics. The development of this field of science is boosted by various domains which are not limited to Cybernetics, Controls, Computers, Mechanics, Bio-Engineering, and Electronics. Among these areas, modelling, control, planning play a fundamental role not only in the growth of industrial robotics, but also towards the advanced fields including healthcare and field robotics.

Through this course the participants will acquire the ability to conduct research, develop innovative designs in the field of systems engineering and control of robots and to direct the development of engineering solutions in new or unfamiliar environments by linking creativity, innovation and transfer of technology.

ABOUT INSTRUCTOR :

Prof. N. Sukavanam received his Ph. D from the Indian Institute of Science, Bangalore in 1985. He served as a Scientist-B at Naval Science and Technological Laboratory, DRDO for two years (1984-86). Then joined as a Research Scientist in the Department of Mathematics, IIT Bombay (1987-90). Worked as a Lecturer at BITS Pilani from 1990 to 1996. In May 1996 he Joined the Department of Mathematics at IIT Roorkee (University of Roorkee at that time) as an Assistant Professor. Currently he is a Professor in the Department of Mathematics IIT Roorkee and Head of the Mathematics from Feb. 2018. His areas of research includes Nonlinear Analysis, Control Theory and Robotics. Professor Sukavanam has published about 80 papers in refereed journals, 30 papers in International Conference Proceedings. He has guided 19 Ph. Ds, 60 M. Sc./M. Phil/MCA Dissertations. Organized International Workshop on Industrial Problems. Developed Pedagogy online course on Mathematics I, offered NPTEL online video course on Dynamical Systems and Control and conducted more than six QIP/Continuing Education courses on Robotics and Control.

Dr. M. Felix Orlando received his Ph.D. from Electrical Engineering Department at Indian Institute of Technology Kanpur (IITK) in 2013, where his advisors were Prof. Laxmidhar Behera, Prof. Ashish Dutta, Prof. Anupam Saxena. In 2015, he completed his post doctoral fellowship at Case Western Reserve University, USA, working with Prof. Tarun Podder, Prof. Yan Yu, Prof. Hutapea focussing on Medical robotics. Dr. Felix Orlando has started as an Assistant Professor in the Department of Electrical Engineering at the Indian Institute of Technology Roorkee (IITR) from November 2015 onwards. He is also the member of IEEE, IEEE-Robotics and Automatin Society (IEEE-RAS), ASME. His current research focuses on medical robotics, rehabilitation robotics, visual servoing and Biomechanics. He is currently the board member of the Student Technical Committee (STC) of robotics, IITR. He has also received the Faculty Initiation Grant (FIG) from IITR for the duration 2016 to 2018. He has received the Early Career Research Award in 2017. He has also coordinated a GIAN course on robotics with Prof. Doik Kim of South Korea in 2017. He has chaired several technical sessions of IEEE international conferences which include, IEEE-AIM 2018, Auckland, IEEE-INDICON 2017, IIT Roorkee, IEEE IECON 2017, Beijing and IEEE ICCSCE 2016, Malaysia. He has presented research papers at various international conferences and has several international robotics journal papers.

COURSE PLAN :

Week 1: Simple manipulators: Two /three arm manipulators and their kinematics equations, Work space Homogeneous Transformation: Rotation, Translation, Composition of homogeneous transformations

Week 2: Denavit-Hartenberg Algorithm: D-H procedure for fixing joint coordinate frames, Robot parameters, Arm matrix, Inverse Kinematics for PUMA, SCARA manipulators.

Week 3: Introduction to Robotic Exoskeletons, Optimal Design of a Three Finger Exoskeleton for Rehabilitation Purpose

Week 4: Differential transformation and velocity of a frame: Derivative of a frame, Velocity, Jacobian, Inverse Jacobian, Trajectory Planning: Polynomial trajectory, Biped trajectory

Week 5: Dynamics: Lagrangian method, Robot dynamics equation, Control: Robot dynamics equation as a control system, Trajectory tracking control, PD controller, Neural network control design

Week 6: Redundancy Resolution of Human Fingers using Robotic Principles, Manipulability Analysis of Human Fingers during Coordinated Object Rotation, Kinematics of Flexible Link Robots,

Week 7: Robot Assisted Needling System for Percutaneous Intervention-An Introduction, Smart Robotic Needles for Percutaneous Cancerous Interventions

Week 8: Robust Force Control of a Two Finger Exoskeleton during Grasping , Neural Control of an Index Finger Exoskeleton – Lecture 1, Neural Control of an Index Finger Exoskeleton – Lecture 2

MOOC APPROVAL PROPOSAL

KTU Course Name	Natural Language Processing
Course Category	Program Elective 6
Course Code	PECST862
MOOC Course Name	Natural Language Processing
NPTEL Subject ID	106105158
Course ID	noc26-cs45
Course URL	https://nptel.ac.in/courses/106105158
COORDINATORS	Prof. Pawan Goyal
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S8

NATURAL LANGUAGE PROCESSING

(Common to CS/CA/CD)

Course Code	PECST862	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of natural language processing (NLP) and language models, focusing on the principles and techniques of prompt engineering to effectively guide and optimize AI-driven outputs.
2. practical skills necessary to design, implement, and evaluate prompt engineering strategies across various applications, while considering the ethical implications and challenges associated with AI-generated content.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to NLP: Introduction to Natural Language Processing - Various stages of traditional NLP – Challenges - Basic Text Processing techniques - Common NLP Tasks. N-gram Language Models - Naive Bayes for Text Classification, and Sentiment Analysis – Evaluation-Precision, Recall and F-measure-Test sets and cross validation.	7
2	Traditional NLP Techniques: Annotating Linguistic Structures - Context-Free Grammars, Constituency Parsing, Ambiguity, CYK Parsing, Dependency Parsing - Transition-Based Dependency Parsing, Graph-Based Dependency Parsing, Evaluation.	7
3	Neural Networks for NLP: Word representations - Lexical Semantics, Vector Semantics, TF-IDF, Pointwise Mutual Information (PMI), Neural Word embeddings - Word2vec, GloVe, Contextual Word Embeddings. Evaluating Vector Models - Feedforward Neural Networks for Text Classification	10

4	Advanced NLP and Applications: Sequence Modelling - Recurrent Neural Networks, RNNs as Language Models, RNNs for NLP tasks, Stacked and Bidirectional RNN architectures, Recursive Neural Networks, LSTM & GRU, Common RNN NLP Architectures, Encoder-Decoder Model with RNNs, Attention models, Transformers. NLP Applications - Machine Translation, Question Answering and Information Retrieval, Introduction to Large Language Models.	12
----------	--	-----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the foundational concepts of NLP and apply that to do text processing.	K3
CO2	Utilize word representations and evaluate vector models for NLP	K3
CO3	Analyse and implement advanced linguistic annotation and parsing techniques	K4
CO4	Apply advanced sequence modeling techniques using Neural Networks	K3
CO5	Apply NLP techniques in machine translation, question answering, and information retrieval.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3								3		
CO2	3	3			3							
CO3	3	3									3	
CO4	3	3	3		3							
CO5	3	3	3			3						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition	Dan Jurafsky and James H. Martin.	Pearson	2006
2	Introduction to Natural Language Processing	Jacob Eisenstein	MIT Press	2019
3	Natural Language Processing with Transformers	Lewis Tunstall, Leandro von Werra, and Thomas Wolf	O'Reilly	2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep learning for Natural Language Processing	Stephan Raaijmakers	Manning	2022
2	Natural Language Processing with PyTorch	Delip Rao and Brian McMahan	O'Reilly	2019
3	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_cs56



NATURAL LANGUAGE PROCESSING

PROF. PAWAN GOYAL

Department of Computer Science and Engineering

IIT Kharagpur

PRE-REQUISITES : Basic knowledge of probabilities for the lectures and python for programming assignment

INDUSTRIES APPLICABLE TO : Microsoft Research, Google, Adobe, Xerox, Flipkart, Amazon

COURSE OUTLINE :

This course starts with the basics of text processing including basic pre-processing, spelling correction, language modeling, Part-of-Speech tagging, Constituency and Dependency Parsing, Lexical Semantics, distributional Semantics and topic models. Finally, the course also covers some of the most interesting applications of text mining such as entity linking, relation extraction, text summarization, text classification, sentiment analysis and opinion mining.

ABOUT INSTRUCTOR :

Prof. Pawan Goyal is an Assistant Professor at the Department of Computer Science and Engineering, IIT Kharagpur. His research interests include Natural Language Processing, Text Mining, Information Retrieval and Sanskrit Computational Linguistics. He has published around 75 research papers in international conferences and journals. He has published in various top-tier conferences and journals including ACL, NAACL, EMNLP, SIGIR, KDD, CIKM, JCDL, ICWSM, CSCW, Coling, IEEE and ACM transactions. Prior to joining IIT Kharagpur, he received his B. Tech. degree in Electrical Engineering from IIT Kanpur in 2007 and his Ph.D. degree in the faculty of Computing and Engineering from University of Ulster, UK in 2011. He was then a Post Doctoral Fellow at INRIA Paris Rocquencourt.

COURSE PLAN :

Week 1: Introduction and Basic Text Processing

Week 2: Spelling Correction, Language Modeling

Week 3: Advanced smoothing for language modeling, POS tagging

Week 4: Models for Sequential tagging – MaxEnt, CRF

Week 5: Syntax – Constituency Parsing

Week 6: Dependency Parsing

Week 7: Distributional Semantics

Week 8: Lexical Semantics

Week 9: Topic Models

Week 10: Entity Linking, Information Extraction

Week 11: Text Summarization, Text Classification

Week 12: Sentiment Analysis and Opinion Mining

MOOC APPROVAL PROPOSAL

KTU Course Name	CLOUD COMPUTING AND APPLICATIONS
Course Category	Open Elective 3
Course Code	OEERT834
MOOC Course Name	Cloud Computing
NPTEL Subject ID	106105167
Course ID	noc26-cs55
Course URL	https://nptel.ac.in/courses/106105167
COORDINATORS	Prof. Soumya Kanti Ghosh
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S8

CLOUD COMPUTING AND APPLICATIONS

Course Code	OEERT 834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Operating System and Networking	Course Type	Theory

Course Objectives:

1. Understand the fundamentals and architecture of cloud computing, including delivery and deployment models.
2. Develop knowledge and skills in cloud security, SLA management, and risk assessment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cloud Computing Fundamentals: What is Cloud Computing, Essential Characteristics, Architectural Influences Cloud delivery models, The SPI Framework, Cloud Software as a Service (SaaS), Cloud Platform as a Service (PaaS), Cloud Infrastructure as a Service (IaaS), Cloud deployment models, Public Clouds, Community Clouds, Private clouds, Hybrid clouds.	9
2	Understanding cloud architecture- Exploring cloud computing stack. Using virtualization technologies, Load balancing and virtualization, Understanding hypervisors, Understanding machine imaging. Cloud storage- Cloud storage providers, Cloud Computing with the Titans– Google, Amazon. Accessing the cloud: Platforms, Web applications, Web APIs.	9
3	Virtual Machines and Containers, Serverless Computing, Using and Managing Containers:-Container Basics, Docker and the Hub. Agents and Microservices: Microservices and Container Resource Managers, Managing Identity in a swarm, A simple Microservices Example, Amazon EC2 Container Service, Google’s Kubernetes.	9

4	SLA Management in Cloud Computing-Types of SLA, Life cycle of SLA, SLA management in cloud. Cloud Information Security Objectives, Cloud Security Services, Relevant Cloud Security Design Principles, Secure Cloud Software Requirements. Privacy and Compliance Risks, Threats to Infrastructure Data and Access Control, Cloud Service Provider Risks. Cloud computing Security Architecture- architecture considerations, identity management and access control	9
----------	--	----------

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the principles of cloud computing.	K2
CO2	Explain the technologies used in cloud computing and virtualization.	K2
CO3	Describe cloud computing and microservices.	K2
CO4	Describe cloud management and cloud security features.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	-	-	-	-	-	-	-	-	-	2
CO4	3	2	-	-	-	-	-	-	-	-	-	2
CO5	3	2	-	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud Security -A Comprehensive Guide to Secure Cloud Computing	Ronald L Krutz and Russell Dean Vines	Wiley Publishing, Inc.	2010
2	Cloud Computing Bible	Barrie Sosinsky	Wiley Publishing	2011
3	Cloud Computing: A Practical Approach	Anthony T. Velte Toby J. Velte, Robert Elsenpeter	The McGraw-Hill	2010
4	Cloud Computing for Science and Engineering https://cloud4scieng.org/chapters/	Ian Foster and Dennis B.Gannon	The MIT Press	2017
5	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg and Andrzej M. Goscinski	Wiley Publishing	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Distributed and Cloud Computing: From parallel processing to Internet of Things	Kai Hwang, Geoffrey C. Fox, Jack K. Dongarra	Morgan Kaufmann	2013
2	Getting Started with Kubernetes: 2nd Edition	Jonathan Baier	Packt publishers	2 nd , 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://en.wikipedia.org/wiki/Cloud_computing https://nptel.ac.in/courses/106105167
2	https://nptel.ac.in/courses/106105167
3	https://nptel.ac.in/courses/106105167
4	https://nptel.ac.in/courses/106105167



Cloud Computing

Prof. Soumya Kanti Ghosh

Department of Computer Science and Engineering
IIT Kharagpur

INTENDED AUDIENCE : CSE,ECE,EE

PREREQUISITES : Basics of Computer Architecture and Organization, Networking

INDUSTRY SUPPORT : IT industries

COURSE OUTLINE:

Cloud computing is a scalable services consumption and delivery platform that provides on-demand computing service for shared pool of resources, namely servers, storage, networking, software, database, applications etc., over the Internet. It is a model for enabling ubiquitous, on-demand access to a shared pool of configurable computing resources, which can be rapidly provisioned and released with minimal management effort. This course will introduce various aspects of cloud computing, including fundamentals, management issues, security challenges and future research trends. This will help students (both UG and PG levels) and researchers to use and explore the cloud computing platforms.

ABOUT THE INSTRUCTOR:

Prof. Soumya K. Ghosh received the Ph.D. and M.Tech. degrees from Department of Computer Science and Engineering, Indian Institute of Technology (IIT), Kharagpur, India. Presently, he is a Professor with Department of Computer Science and Engineering, IIT Kharagpur. Before joining IIT Kharagpur, he worked for the Indian Space Research Organization in the area of satellite remote sensing and geographic information systems. He has more than 200 research papers in reputed journals and conference proceedings. His research interests include spatial data science, spatial web services and cloud computing.

COURSE PLAN:

Week 1: Introduction to Cloud Computing

Week 2: Cloud Computing Architecture

Week 3: Service Management in Cloud Computing

Week 4: Data Management in Cloud Computing

Week 5: Resource Management in Cloud

Week 6: Cloud Security

Week 7: Open Source and Commercial Clouds, Cloud Simulator

Week 8: Research trend in Cloud Computing, Fog Computing

Week 9: VM Resource Allocation, Management and Monitoring

Week 10: Cloud-Fog-Edge enabled Analytics

Week 11: Serverless Computing and FaaS Model

Week 12: Case Studies and Recent Advancements

MOOC APPROVAL PROPOSAL

KTU Course Name	PLC & Automation
Course Category	Open Elective 3
Course Code	OEEET832
MOOC Course Name	Industrial Automation and Control
NPTEL Subject ID	108105088
Course ID	noc26-ee47
Course URL	https://nptel.ac.in/courses/108105088
COORDINATORS	Prof. Alokanti Deb
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S8

PLC AND AUTOMATION

Course Code	OEEET832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Learn the roles, architectures, and interfacing techniques of computer-based measurement and control systems, including HMI and hardware integration.
2. Gain hands-on experience with PLC programming and simulation, and understand the functionalities and interfacing of Distributed Control Systems for process control.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to computer based control system -Role of computers in measurement and (process) control Basic components of computer based measurement and control systems Architecture – computer based process control system –Centralised, Distributed and Hierarchical. Human Machine Interface (HMI) Hardware for computer based process control system, Interfacing computer system with process. Architecture of DDC, SCADA and DCS. Programmable logic Controller (PLC): Introduction, Evolution, Relay VS PLC VS Computer	9
2	PLC- Hardware and Internal Architecture-Input –output devices .Basics of Ladder Programming, on/off instructions, internal relay, jump instructions, data handling instruction, data manipulation instructions, Arithmetic and Comparison ,PID and other important instructions	9
3	Timers and Counters in PLC. Problems. Design Development and Simulation of PLC Programme Program on Temperature control Valve sequencing, Conveyor belt control and Control of a process.	9

	PLC Installation, trouble shooting and maintenance, Design of Alarms and Interlocks, Networks of PLC Distributed Control System- DCS - Evolution– Various Architectures – Comparison – Local control unit	
4	DCS -LCU Languages-Process interfacing issues-communication facilities- Operator interface-Low level and High level Operator interface- Displays - Engineering interfaces – Low level and high level engineering interfaces – Factors to be considered in selecting DCS – Other key issues in DCS – Packaging and Power system issues.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Micro projects on automation using PLC and DCS for student group comprising of 3 students.

Report – 5 marks

Working Model – 15 Marks

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic architecture and components of computer-based measurement and control systems.	K2
CO2	Understand the human-machine interfaces (HMI) and learn the hardware and interfacing techniques needed to integrate computer systems with process controls.	K2
CO3	Create and troubleshoot PLC programs using ladder logic for various applications.	K5
CO4	Understand and apply the architecture and interfaces of Distributed Control Systems in various process control settings.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3				2							
CO4	3											

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineer's Handbook – Process Control,	B G Liptak	CRC Press	4 th edition
2	Understanding Distributed Processor Systems for Control,	Samel M. Herb	ISA Publication	1 st edition 1999
3	Programmable Logic Controllers – Principles and Applications.	John W. Webb & Ronald A. Reiss,	PHI	5 th edition
4	Computer Control of Processes,	M. Chidambaram	Alpha Science International Ltd	1 st edition 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Software and Digital Networks, CRC Press.	B G Liptak	CRC	3 rd edition
2	Programmable Logic Controllers – Programming Methods and Applications, Pearson Education.	John R. Hackworth & Frederick D. Hackworth Jr	Pearson	1 st edition 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me67/preview
2	https://onlinecourses.nptel.ac.in/noc21_me67/preview
3	https://onlinecourses.nptel.ac.in/noc21_me67/preview
4	https://onlinecourses.nptel.ac.in/noc21_me67/preview



INDUSTRIAL AUTOMATION AND CONTROL

PROF. ALOKKANTI DEB

Department of Electrical Engineering,
IIT Kharagpur

PRE-REQUISITES : Electrical Networks, Control Systems

INTENDED AUDIENCE : Any interested student

INDUSTRIES APPLICABLE TO : All Process Control (Oil and Gas, Chemical), Manufacturing (Machine tools, Textile) etc.

COURSE OUTLINE :

This course provides an overall exposure to the technology of Industrial Automation and Control as widely seen in factories of all types both for discrete and continuous manufacturing. The course, in 52 lectures, discusses a wide range of related topics from the advantage and architecture of automation systems, measurement systems including sensors and signal conditioning, discrete and continuous variable control systems, hydraulic, pneumatic and electric actuators, industrial communication and embedded computing and CNC Machines. A student of IIT Kharagpur once commented - “because of the course I can identify and relate to much of the equipment that I see in a factory”.

ABOUT INSTRUCTOR :

Prof. Alok Kanti Deb received the B.E. (Hons) degree in Electrical Engineering from the Bengal Engineering College, Calcutta University, Howrah, India, and the M.Tech. (Control Engg. and Instrumentation) and Ph.D degrees in electrical engineering from IIT Delhi, Delhi., India in 1994, 1999 and 2006 respectively. He is currently a Professor with the Department of Electrical Engineering, IIT Kharagpur, Kharagpur, India. He has taught several UG and PG courses and instructed their associated labs, sessionals and seminars in the Electrical Engineering Department like, Electrical Technology, Embedded Systems, Control System Engineering, Estimation of Signals and Systems, Intelligent Control, Industrial Automation and Control, Control Theory and Digital Signal Processing. He has also taught several interdisciplinary courses like Instrumentation and Control (SMST) and Automation and Control (Steel Technology Center). His research interests include control systems, computational intelligence and automotive diagnostics. He has completed research projects with General Motors, USA and Department of Electronics and Information Technology (DeiTY), Govt of India. He is presently involved in research projects with Ministry of Human Resource Development (MHRD), Aeronautical Development Agency (ADA), Bangalore, India, Ministry of Railways, Govt of India and UK India Collaboration in Smart Grids and Energy Storage.

He is a mentor to the Kharagpur Robo Soccer Students' Group (KRSSG). He was the Co-ordinator of the workshops, “Embedded and Reconfigurable Computing for Control and Signal Processing & Xilinx Embedded Design flow using Zynq and Vivado Design suite”, ATDC, IIT Kharagpur, Jan 6-10, 2014 and “Estimation and Control: Advanced Theory and Applications”, Dept of Electrical Engg., IIT Kharagpur, Dec 25-30, 2009. He was a organizing committee member of ICIIS-2008 & ICPS2009 at IIT Kharagpur. He is the Chair, IEEE Control System Society, IEEE Kharagpur Section.

He was an Assistant Professor with the Centre for Soft Computing Research, Indian Statistical Institute, Kolkata, India from 2005 to 2007. He served as an Engineer with Calcutta Electric Supply Corporation Ltd., Kolkata, from 1994 to 1997, and was involved in commissioning, maintenance and condition monitoring of turbine-generator sets of various ratings.

Prof. Deb received the Student Travel Award from the IEEE Neural Network Society for attending the IEEE World Congress on Computational Intelligence in 2002. He also received Travel Award from the IEEE Computational Intelligence Society in 2014 to attend IEEE Symposium Series on Computational Intelligence (SSCI-2014). He has published several papers in international journals and international conferences, 1 book chapter and co-authored the book, “Industrial Instrumentation, Control and Automation”, published by Jaico, Mumbai, 2013. He is the holder of the patent, “STATE ESTIMATION, DIAGNOSIS AND CONTROL USING EQUIVALENT TIME SAMPLING” (US Patent No. – 8,751,097 B2, dt, June 10, 2014). He regularly reviews papers from several journals and conferences.

COURSE PLAN:

Module I

- Introduction
- Introduction(Cont.)
- Architecture of Industrial Automation Systems
- Architecture of Industrial Automation Systems(Cont.)

Module II

- Measurement Systems Characteristics
- Measurement Systems Characteristics(Cont.)
- Data Acquisition Systems
- Data Acquisition Systems(Cont.)

Module III

- Introduction to Automatic Control
- Introduction to Automatic Control(Cont.)
- P-I-D Control
- P-I-D Control(Cont.)
- PID Control Tuning
- PID Control Tuning(Cont.)
- Feedforward Control Ratio Control
- Feedforward Control Ratio Control(Cont.)
- Time Delay Systems and Inverse Response Systems
- Time Delay Systems and Inverse Response Systems(Cont.)
- Special Control Structures
- Special Control Structures(Cont.)
- Concluding Lesson on Process Control (Self-study)
- Introduction to Sequence Control, PLC , RLL
- Introduction to Sequence Control, PLC , RLL(Cont.)
- Sequence Control. Scan Cycle, Simple RLL Programs
- Sequence Control. Scan Cycle, Simple RLL Programs(Cont.)
- Sequence Control. More RLL Elements, RLL Syntax
- Sequence Control. More RLL Elements, RLL Syntax(Cont.)
- A Structured Design Approach to Sequence Control
- A Structured Design Approach to Sequence Control(Cont.)
- PLC Hardware Environment
- PLC Hardware Environment(Cont.)

Module IV

- Flow Control Valves
- Flow Control Valves(Cont.)
- Hydraulic Control Systems - I
- Hydraulic Control Systems - I(Cont.)
- Hydraulic Control Systems - II
- Hydraulic Control Systems - II(Cont.)
- Industrial Hydraulic Circuit
- Industrial Hydraulic Circuit(Cont.)
- Pneumatic Control Systems - I
- Pneumatic Control Systems - I(Cont.)
- Pneumatic Systems - II
- Pneumatic Systems - II(Cont.)
- Energy Savings with Variable Speed Drives
- Energy Savings with Variable Speed Drives(Cont.)
- Introduction To CNC Machines
- Introduction To CNC Machines(Cont.)

Module V

- The Fieldbus Network - I
- The Fieldbus Network - I(Cont.)
- Higher Level Automation Systems
- Higher Level Automation Systems(Cont.)
- Course Review and Conclusion (Self-study)

MOOC APPROVAL PROPOSAL

KTU Course Name	Internet of Things
Course Category	Open Elective 3
Course Code	OEECT831
MOOC Course Name	Introduction to Internet Of Things
NPTEL Subject ID	106105166
Course ID	noc26-cs37
Course URL	https://nptel.ac.in/courses/106105166
COORDINATORS	Prof. Sudip Misra
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER S8

INTERNET OF THINGS

Course Code	OEECT 831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. This course aims to introduce IoT fundamentals.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT technology: Definitions of IoT, Characteristics of IoT devices – power, computational constraints, IoT Architectural view – Middleware based architecture, Service oriented architecture, M2M Communication and IoT, Typical application areas of IoT technology (case studies of at least four domains) - Energy management and Smart grid, IoT for Home, Cities, Environment monitoring, Agriculture, Supply chain and customer monitoring	9
2	Components of IoT technology: Identification/Addressing - Electronic Product Codes, RFID, ubiquitous code, IPv4, IPv6. Sensors and Actuators*. IoT Hardware**, IoT Software – overview of Operating systems, Firmware, Middle ware, Application software used in IoT. Connectivity for IoT devices – characteristics.	9
3	Communication technologies for IoT : Zigbee - key features, architecture, limitations, Bluetooth technology - bluetooth stack, piconet, scatternet, limitations, Bluetooth Low Energy (key features, architecture, limitations), Wifi (IEEE 802.11) technology – key features, limitations, Cellular	9

	technology – GSM, 3G, 4GLTE (overview), features, limitations, LoRa technology – features, LoRaWAN architecture, 6LoWPAN – features, protocol stack, Narrow Band (NB- IoT) – features, applications, Sigfox – features, applications	
4	IoT Data Management : Storage technologies for IoT hardware – Volatile, Non-volatile, Embedded (MTP/OTP), external flash (NAND/NOR), DRAM, eflash, UFS, eMMC (overview of technologies). Cloud and IoT, Cloud computing – architecture, advantages of cloud computing, Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS). Case study of commercial cloud computing platforms like - Microsoft Azure IoT Suite, Google Cloud's IoT Platform, IBM Watson IoT Platform. IoT analytics	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain in a concise manner the architecture of IoT	K2
CO2	Identify various hardware and software components used in IoT	K3
CO3	Discuss the various communication technologies and interfaces in IoT	K2
CO4	Describe the usage of modern technologies like cloud computing for data management in IoT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								2
CO2	3	2	2	2								2
CO3	3	2	2	1								2
CO4	3	2	2	1								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation



INTRODUCTION TO INTERNET OF THINGS

PROF. SUDIP MISRA

Department of Computer Science and Engineering
IIT Kharagpur

PRE-REQUISITES : Basic programming knowledge

INTENDED AUDIENCE : CSE, IT, ECE, EE, Instrumentation Engineering, Industrial Engineering

COURSE OUTLINE :

Internet of Things (IoT) is presently a hot technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT. IoT cuts across different application domain verticals ranging from civilian to defence sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support IoT. Today it is possible to envision pervasive connectivity, storage, and computation, which, in turn, gives rise to building different IoT solutions. IoT-based applications such as innovative shopping system, infrastructure management in both urban and rural areas, remote health monitoring and emergency notification systems, and transportation systems, are gradually relying on IoT based systems. Therefore, it is very important to learn the fundamentals of this emerging technology.

ABOUT INSTRUCTOR :

Prof. Sudip Misra is a Professor in the Department of Computer Science and Engineering at the Indian Institute of Technology Kharagpur. Prior to this he was associated with Cornell University (USA), Yale University (USA), Nortel Networks (Canada) and the Government of Ontario (Canada). He received his Ph.D. degree in Computer Science from Carleton University, in Ottawa, Canada. He has several years of experience working in the academia, government, and the private sectors in research, teaching, consulting, project management, architecture, software design and product engineering roles. His current research interests include Wireless Ad Hoc and Sensor Networks, Internet of Things (IoT), Computer Networks, Learning Systems, and algorithm design for emerging communication networks. Dr. Misra is the author of over 260 scholarly research papers, including 140+ reputed journal papers. He has won seven research paper awards in different conferences. Recently, he and his students won Samsung Innovation Award and the IEEE ComSoc Student Competition. He was awarded the fellow of NASI. He was also awarded the IEEE ComSoc Asia Pacific Outstanding Young Researcher Award at IEEE GLOBECOM 2012, Anaheim, California, USA. He was also the recipient of several academic awards and fellowships such as the Young Scientist Award (National Academy of Sciences, India), Young Systems Scientist Award (Systems Society of India), Young Engineers Award (Institution of Engineers, India), (Canadian) Governor General's Academic Gold Medal at Carleton University, the University Outstanding Graduate Student Award in the Doctoral level at Carleton University and the National Academy of Sciences, India - Swarna Jayanti Puraskar (Golden Jubilee Award). Dr. Misra was also awarded the Canadian Government's prestigious NSERC Post-Doctoral Fellowship and the Humboldt Research Fellowship in Germany. Dr. Misra has been serving the editorial boards of distinguished journals such as the Transactions on Vehicular Technology, Transactions on Mobile Computing, International Journal of Communication Systems (Wiley) and the IET Wireless Sensor Systems (UK). In the past, he served as the Associate Editor/Editorial Board Member of the Telecommunication Systems Journal (Springer), Security and Communication Networks Journal (Wiley), and the EURASIP Journal of Wireless Communications and Networking, IET Communications Journal, and the Computers and Electrical Engineering Journal (Elsevier). Dr. Misra has published 10 books in the areas of wireless ad hoc networks, wireless sensor networks, wireless mesh networks, communication networks and distributed systems, network reliability and fault tolerance, and information and coding theory, published by reputed publishers such as Cambridge University Press, Springer, Wiley, and World Scientific.

COURSE PLAN :

Week 1: Introduction to IoT: Part I, Part II, Sensing, Actuation, Basics of Networking: Part-I

Week 2: Basics of Networking: Part-II, Part III, Part IV, Communication Protocols: Part I, Part II

Week 3: Communication Protocols: Part III, Part IV, Part V, Sensor Networks: Part I, Part II

Week 4: Sensor Networks: Part III, Part IV, Part V, Part VI, Machine-to-Machine Communications

Week 5: Interoperability in IoT, Introduction to Arduino Programming: Part I, Part II, Integration of Sensors and Actuators with Arduino: Part I, Part II

Week 6: Introduction to Python programming, Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi

Week 7: Implementation of IoT with Raspberry Pi (contd), Introduction to SDN, SDN for IoT

Week 8: SDN for IoT (contd), Data Handling and Analytics, Cloud Computing

Week 9: Cloud Computing(contd), Sensor-Cloud

Week 10: Fog Computing, Smart Cities and Smart Homes

Week 11: Connected Vehicles, Smart Grid, Industrial IoT

Week 12: Industrial IoT (contd), Case Study: Agriculture, Healthcare, Activity Monitoring

MOOC APPROVAL PROPOSAL

KTU Course Name	Organizational Behavior and Business Communication
Course Category	HMC Elective
Course Code	UEHUT803
MOOC Course Name	Organizational Behaviour: Individual Dynamics in Organization
NPTEL Subject ID	110103433
Course ID	noc26-mg20
Course URL	https://nptel.ac.in/courses/110103433
COORDINATORS	Prof. Abraham Cyril; Issac
Offering Institute	IIT Guwahati
Duration	12 weeks
Platform	NPTEL

SEMESTER S8

**ORGANIZATIONAL BEHAVIOUR AND BUSINESS
COMMUNICATION**

Course Code	UEHUT803	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students to the fundamentals of Organizational Behaviour (OB) and Business communication
2. To become familiar with working with people, nature of organizations, communication, leadership and motivation of people.
3. To help students develop a conceptual understanding of OB theories
4. To enable the students to put the ideas and skills of OB into practice

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Organizational Behaviour: Definition, Evolution of the Concept of OB, Contributions to OB by major behavioural science disciplines, Challenge and Opportunities for OB managers, Models of OB study (Custodial Model, Supportive Model, Collegial Model, Human Value Model)	6
2	Individual Behaviour-Attitudes and Job Satisfaction, Components of Attitude, Personality and Values, Personality Determinants, Types of Values, Perception, Factors influencing perception. Group Behaviour, Formation of Group, Group decision making, creating effective teams, Culture, Culture's function	6
3	Motivation and Leadership Concept of motivation, Definition, Theories of Motivation, Maslow's need Theory, Theory X and Theory Y, Two Factor	6

	Theory, Vroom's Expectancy Theory. Concept of Leadership, Theories of leadership, Traits of good Leader, Difference between Leader and Manager	
4	Business communication- Functions, Types and process of communication- downward, upward communication- Interpersonal and Group Communication- - Choices of Communication Channel-Cross functional communication- Conflict management communication Modern forms of business communication- Global communication	6

Course Assessment Method
(CIE: 50 marks , ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	15	15	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
- Minimum 1 and Maximum 2 Questions from each module. - Total of 6 Questions, each carrying 3 marks (6x3 =18marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the models and frameworks of organizational behaviour and business communication	K2
CO2	Develop the team working and decision-making capacity by assessing the influence of Individual and Group Behaviour towards the organizational Goal	K3
CO3	Apply and analyse the Theories of motivation and leadership in an organisation	K3 & K4
CO4	Explore the business communication methods for organizational effectiveness	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1									3	2	2	
CO2									3	2	2	
CO3									3	2	2	
CO4									3	2	2	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organizational Behaviour	Kavita Singh	Vikas Publications	3e-2015
2	Organizational Behavior	Robbins, Timothy Judge, Seema Sanghi	Pearson	18e-2018
3	Organizational Behavior: Improving Performance and Commitment in the Workplace	Jason Colquitt, Jeffery A. LePine, and Michael J. Wesson	McGraw-Hill Education	2018
4	Business Communication: Essential Strategies for 21st Century Managers	Shalini Verma	Vikas Publishing House	2e-2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organizational Behaviour- An Evidence-Based Approach	Fred Luthans, Brett C. Luthans, Kyle W. Luthans	McGraw Hill Inc	13 ^e -2015
2	Organizational Behaviour	John Newstrom and Keith Davis	Tata McGraw Hill	11 ^e - 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	www.algonquincollege.com/ccol/courses/organizational-behaviour/
2	www.algonquincollege.com/ccol/courses/organizational-behaviour/
3	www.algonquincollege.com/ccol/courses/organizational-behaviour-3
4	https://www.imi.edu/delhi/organizational_behaviour_human_resources



ORGANIZATIONAL BEHAVIOUR: INDIVIDUAL DYNAMICS IN ORGANIZATION

PROF. ABRAHAM CYRIL ISSAC

Department of Management
IIT Guwahati

INTENDED AUDIENCE : MBA, EMBA, Psychology students pursuing or have completed post-graduate course; students of any other discipline pursuing or have completed masters

INDUSTRY SUPPORT : Any organization with significant human interaction within the workforce will value this course as it covers the basic tenets of individual behaviour within the organization

COURSE OUTLINE : Organizational behaviour is an essential part of management studies as it centers around the activities of individuals nested within the system and interacting amongst themselves and the system to create output. The focus on individual dynamics is the crux of this course which will equip the course takers to understand the theories and applications of behavior in work settings. The course encompasses individual attributes which impact workers' behavior at their workplace. Individuals are the basic blocks of organizational behavior. Some individual characteristics remain the same across time whereas some change depending on their internal and external conditions. By the end of this course, students are expected to have a thorough understanding of the individual dynamics that play an important role in organizational context. They will be able to i) Analyze the human factors responsible for various events occurring within the organization ii) Think critically about situations using theoretical underpinnings iii) Undertake better management practices and decisions.

ABOUT INSTRUCTOR : Prof. Abraham Cyril Issac is an Assistant Professor in the School of Business at the Indian Institute of Technology Guwahati, India. He did Ph.D. from Indian Institute of Technology, Madras, and Swinburne University of Technology, Melbourne. He is a Mechanical engineer from Malaviya National Institute of Technology Jaipur and has completed postgraduate degrees in Business Administration, Psychology, and Business Law from the National Institute of Technology Calicut, University of Madras, and NLSIU Bangalore respectively. He is a trained thermal powerplant engineer. Now he specializes in Organizational Behavior with a special focus on a relatively new construct of Knowledge hiding and is one of the most productive researchers in the area. He is a recipient of the IIT Madras Institute Research (IR) Award 2021 awarded by the institute for excellent Ph.D work and the R. N. Rajendran Award for Best Thesis in the area of Organizational Behaviour/Human Resource Management. His academic interests include Organizational Behaviour, Human Resource Management, Knowledge Management, and Macroeconomics. Apart from his regular teaching and research assignments, he also engages actively in consultancy, training and coaching programmes.

COURSE PLAN :

Week 1: Understanding organizational behaviour

Lecture 1: Understanding organizational behavior: Tracing the evolution-1

Lecture 2: Understanding organizational behavior: Tracing the evolution-2

Lecture 3: Individual in the organization: The building blocks

Lecture 4: Understanding individual differences-1

Lecture 5: Understanding individual differences -2

Week 2: Diversity and inclusion in organization

Lecture 1: Diverse workforce, inclusive mindset- 1

Lecture 2: Diverse workforce, inclusive mindset- 2

Lecture 3: Perception of diversity and inclusion

Lecture 4: Ableism and inclusion

Lecture 5: Diversity management

Week 3: Perception and decision making

Lecture 1: The perceptual process

Lecture 2: Factors that influence perception

Lecture 3: Perception and decision making

Lecture 4: What affects decisions?

Lecture 5: Ethical decision making

Week 4: Affect and emotions

Lecture 1: Affect and emotions

Lecture 2: Affective events theory

Lecture 3: emotional intelligence

Lecture 4: Understanding stress

Lecture 5: Emotions and moods: Application at workplace

Week 5: Personality

Lecture 1: Understanding self and personality

Lecture 2: Types and theories of personality

Lecture 3: Measuring personality

Lecture 4: Personality traits relevant to organization

Lecture 5: Assessing personality: caveats and concerns

Week 6: Values at workplace

Lecture 1: Values and its importance

Lecture 2: Sources and types of values

Lecture 3: Values that cut across cultures

Lecture 4: Person-job fit

Lecture 5: Person-organization fit

Week 7 : Motivational differences in individuals

Lecture 1: Motivation: Basic understanding and definition

Lecture 2: Tracing the roots: Early theories-1

Lecture 3: Tracing the roots: Early theories-1

Lecture 4: Keeping up with times: Contemporary theories- 1

Lecture 5: Keeping up with times: Contemporary theories- 2

Week 8: Motivation: Application at workplace

Lecture 1: Job Design and job characteristics model

Lecture 2: Employee involvement

Lecture 3: Motivating the employees: Strategies for organization-1

Lecture 4: Motivating the employees: Strategies for organization-2

Lecture 5: Organizational justice and employee motivation

Week 9: Learning in organization

Lecture 1: Defining learning

Lecture 2: How do individuals learn?

Lecture 3: Theories of learning- 1

Lecture 4: Theories of learning- 2

Lecture 5: Learning in an organizational setting

Week 10: Creativity, psychological capital and mindfulness

Lecture 1: Understanding creativity

Lecture 2: Stages of individual creativity

Lecture 3: Creativity and problem solving

Lecture 4: Defining psychological capital

Lecture 5: Using Psychological capital and mindfulness at work

Week 11: Knowledge sharing and hiding

Lecture 1: Understanding knowledge sharing

Lecture 2: Understanding knowledge hiding- what it is and what it is not

Lecture 3: Individual factors affecting knowledge hiding

Lecture 4: Integrating knowledge sharing and hiding behavior

Lecture 5: Strategies for individuals to foster knowledge sharing

Week 12: Employee voice and silence

Lecture 1: Understanding employee voice and silence

Lecture 2: Individual factors affecting voice behavior

Lecture 3: Individual factors affecting silence

Lecture 4: Can silence be strategic?

Lecture 5: Strategies for fostering safe environment at work

MOOC APPROVAL PROPOSAL

KTU Course Name	Project Management: Planning, Execution, Evaluation and Control
Course Category	HMC Elective
Course Code	UEHUT704
MOOC Course Name	Project Management : Planning, Execution, Evaluation And Control
NPTEL Subject ID	110105167
COORDINATORS	Prof. Sanjib Chowdhury
Offering Institute	IIT Kharagpur
Duration	8 Weeks
Platform	-

SEMESTER S7

PROJECT MANAGEMENT : PLANNING, EXECUTION,EVALUATION AND CONTROL

Course Code	UEHUT704	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand and apply project management concepts and tools, including project planning, scheduling, and resource allocation, to manage project scope, cost, and time effectively.
2. Develop practical skills in project execution, risk management, and the use of project management software to ensure successful project evaluation, performance tracking, and closure.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Project Management:- Concept of a project-importance and challenges- classification of projects; Project management essentials-importance of project management-attributes of a good project manager-comparison: project management vs. general management; The project life cycle-phases of project life cycle; Establishing project priorities (scope, cost, time); Work break down structure; Capital budgeting process-planning-analysis- selection- financing-implementation-review.	6
2	Project Initiation and Planning:- Generation and screening of project ideas; Market and demand analysis-demand forecasting techniques-Qualitative and Quantitative; Market planning and marketing research process; Technical and financial analysis; Forms of Project organization; Human aspects of project management.	6

3	Project Scheduling and Resource Management:- Project time and cost estimation; Time Management- Developing Project Plan -Network Analysis using PERT/ CPM technique; Resource Management and Cost Management: Resource levelling- Scheduling and allocating project resources and costs.	6
4	Advanced Project Management and Closure:- Reducing Project duration - crashing project activities to speed up a project; Project Risk Management; Project Evaluation; Project progress and performance Management; Project closure and oversight; Introduction to various Project management software.	6

Course Assessment Method
(CIE: 50 marks , ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	15	15	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
- Minimum 1 and Maximum 2 Questions from each module. - Total of 6 Questions, each carrying 3 marks (6x3 =18marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the essential concepts of project management, including project classification, the project life cycle, and the capital budgeting process.	K2
CO2	Perform market and demand analyses, apply forecasting techniques, and carry out technical and financial assessments for project viability.	K3
CO3	Develop detailed project schedules using PERT/CPM techniques and effectively manage project resources and costs.	K3
CO4	Apply methods for reducing project duration, manage project risks, and utilize project management software for project evaluation and closure.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1							1		2	
CO2	3	1							1		2	
CO3	3	2							1		2	
CO4	3	2			1				1		2	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Projects Planning, Analysis, Selection, Implementation and Review	Prasanna Chandra	Tata McGraw –Hill	10 th Edition, 2024
2	Project Management: The Managerial Process	Erik W. Larson and Clifford F. Gray	McGraw-Hill Education	8th Edition, 2020
3	Project Management	Dennis Lock	Gower	10 th Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Management for Engineering, Business, and Technology	John M. Nicholas and Herman Steyn	Routledge	6th Edition, 2021
2	Text book of Project management	Gopalakrishnan P & Ramamoorthy V E	Laxmi Publications	1 st Edition, 2022
3	Fundamentals of Project Management	K. Nagarajan	New Age International Publishers	6th Edition, 2019
4	Production and Operations management	R. Panneerselvam	PHI Learning Pvt. Ltd	3 rd Edition, 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=obzp6biyAN0
2	https://www.youtube.com/watch?v=dAJy-3A6lPM
3	https://onlinecourses.nptel.ac.in/noc24_mg01
4	https://onlinecourses.nptel.ac.in/noc24_mg01



PROJECT MANAGEMENT : PLANNING, EXECUTION, EVALUATION AND CONTROL

PROF. SANJIB CHOWDHURY

Department of Economics and Strategy
IIT Kharagpur

PRE-REQUISITES : Must be a graduate in management, engineering, science, commerce, or equivalent degree

INTENDED AUDIENCE : Management and Engineering students (MBA, BE/ B.Tech, ME/ M.Tech), academics, practicing managers and professionals at various levels; Commerce, Cost and Chartered Accountancy, and related disciplines.

INDUSTRY SUPPORT : All organizations be it Government or non- Government, profit or non-profit, business or charity, manufacturing or service, utility or IT, military and others need Project Management for planning, financing, implementation, evaluation, and control of projects to accomplish their long- term goals and objectives of their organizations.

COURSE OUTLINE :

Projects are the modus operandi of implementing organizational strategy and achieving strategic goals and objectives. Its importance is well-acknowledged and pervasive in the business world such as construction, engineering, mining, hi-tech, electronics, logistics, defense forces, product development, R&D, and so on. Project management has developed to the point where it is a professional discipline having its own body of knowledge and skills. In order to achieve business excellence, it is necessary to develop expertise in project planning, executing, evaluating, and managing projects. The course would help the academics as well as practicing managers and professionals to learn modern project management tools and techniques, scheduling of activities, resources and cost; and enhance their skills in expediting projects by prudently crashing certain activities. It would also provide them a holistic view of project management covering among others project risk and quality management, procurement and contract management, communication management, developing leadership skills and managing PM team, project performance evaluation, project audit and closure. The course would help the participants to apply Project Management knowledge in real life situation.

ABOUT INSTRUCTOR :

Prof. Sanjib Chowdhury is a Visiting Professor (full-time) at the Indian Institute of Technology Kharagpur, specializing in Strategic Management, and Operations Management. He brings an impressive 41 years of professional experience, including 34 years in the oil and gas industry and 7 years in academia. His industry experience includes two major oil companies in India and Kuwait in the areas of Strategic and Corporate Planning, Business Performance Improvement, Operations and Process Optimization, Project Management, and HR. He is the recipient of "Kuwait Gulf Oil Company's Ideal Employee Award", "ONGC Chairman's Award", "ONGC Director's Award" to mention a few, and has held several responsible positions exhibiting his leadership and technical expertise.

Based on his work experience, he has solely authored a book titled "Optimization and Business Improvement Studies in Upstream Oil and Gas Industry", published by John Wiley & Sons Inc., NJ, USA which reflects his deep understanding and knowledge of the industry and is a valuable resource for professionals and academia. Dr. Chowdhury received B.Tech. (Hons.), M.Tech., and Ph.D. in Industrial Engineering and Management all from IIT Kharagpur. He has also published several technical papers in esteemed journals, further contributing to his field.

COURSE PLAN :

Week 1: Introduction to Project Management, and Project Life Cycle Management; Organization Strategy and Project Selection.

Week 2: Project Management Organization Structure and Organization Culture; Defining Project: Definition, Activities, Work Breakdown Structure etc.

Week 3: Project Time and Cost Estimation; Developing Project Plan: Network Analysis using PERT/ CPM Techniques.

Week 4: Project Risk Management - Identification, Quantification, and Mitigation of Risks; Risk Assessment of Capital Projects.

Week 5: Resource and Cost Management: Resource Levelling, Scheduling, and Allocation.

Week 6: Reducing Project duration : Activity Crashing; Project Evaluation, Progress and Performance Measurement.

Week 7: Project Outsourcing, Negotiation, and Managing inter-organizational Relations.

Week 8: Contract Management; Project Closure; and Project Oversight.

MOOC APPROVAL PROPOSAL

KTU Course Name	Project Management: Planning, Execution, Evaluation and Control
Course Category	HMC Elective
Course Code	UEHUT704
MOOC Course Name	Project Management
NPTEL Subject ID	110107430
Course ID	noc26_mg77
Course URL	https://nptel.ac.in/courses/110107430
COORDINATORS	Prof. A. Ramesh
Offering Institute	IIT Roorkee
Duration	8 Weeks
Platform	NPTEL

SEMESTER S7

PROJECT MANAGEMENT : PLANNING, EXECUTION,EVALUATION AND CONTROL

Course Code	UEHUT704	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand and apply project management concepts and tools, including project planning, scheduling, and resource allocation, to manage project scope, cost, and time effectively.
2. Develop practical skills in project execution, risk management, and the use of project management software to ensure successful project evaluation, performance tracking, and closure.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Project Management:- Concept of a project-importance and challenges- classification of projects; Project management essentials-importance of project management-attributes of a good project manager-comparison: project management vs. general management; The project life cycle-phases of project life cycle; Establishing project priorities (scope, cost, time); Work break down structure; Capital budgeting process-planning-analysis- selection- financing-implementation-review.	6
2	Project Initiation and Planning:- Generation and screening of project ideas; Market and demand analysis-demand forecasting techniques-Qualitative and Quantitative; Market planning and marketing research process; Technical and financial analysis; Forms of Project organization; Human aspects of project management.	6

3	Project Scheduling and Resource Management:- Project time and cost estimation; Time Management- Developing Project Plan -Network Analysis using PERT/ CPM technique; Resource Management and Cost Management: Resource levelling- Scheduling and allocating project resources and costs.	6
4	Advanced Project Management and Closure:- Reducing Project duration - crashing project activities to speed up a project; Project Risk Management; Project Evaluation; Project progress and performance Management; Project closure and oversight; Introduction to various Project management software.	6

Course Assessment Method
(CIE: 50 marks , ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	15	15	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the essential concepts of project management, including project classification, the project life cycle, and the capital budgeting process.	K2
CO2	Perform market and demand analyses, apply forecasting techniques, and carry out technical and financial assessments for project viability.	K3
CO3	Develop detailed project schedules using PERT/CPM techniques and effectively manage project resources and costs.	K3
CO4	Apply methods for reducing project duration, manage project risks, and utilize project management software for project evaluation and closure.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1							1		2	
CO2	3	1							1		2	
CO3	3	2							1		2	
CO4	3	2			1				1		2	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Projects Planning, Analysis, Selection, Implementation and Review	Prasanna Chandra	Tata McGraw –Hill	10 th Edition, 2024
2	Project Management: The Managerial Process	Erik W. Larson and Clifford F. Gray	McGraw-Hill Education	8th Edition, 2020
3	Project Management	Dennis Lock	Gower	10 th Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Management for Engineering, Business, and Technology	John M. Nicholas and Herman Steyn	Routledge	6th Edition, 2021
2	Text book of Project management	Gopalakrishnan P & Ramamoorthy V E	Laxmi Publications	1 st Edition, 2022
3	Fundamentals of Project Management	K. Nagarajan	New Age International Publishers	6th Edition, 2019
4	Production and Operations management	R. Panneerselvam	PHI Learning Pvt. Ltd	3 rd Edition, 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=obzp6biyAN0
2	https://www.youtube.com/watch?v=dAJy-3A6lPM
3	https://onlinecourses.nptel.ac.in/noc24_mg01
4	https://onlinecourses.nptel.ac.in/noc24_mg01



PROJECT MANAGEMENT

PROF. RAMESH ANBANANDAM

Department of Management Studies
IIT Roorkee

INTENDED AUDIENCE : 1. Undergraduate Engineering Courses-All discipline

2. Management Courses- All discipline

INDUSTRY SUPPORT : All software companies.

Manufacturing Companies

Construction companies

COURSE OUTLINE :

Many often question the significance of project management and the necessity of a manager overseeing diverse projects within organisations. A project manager plays a crucial role in unifying a team towards a common goal and ensuring the seamless progress of a project. Project management provides guidance, fosters a sense of direction, propels the team forward, eliminates obstacles, coaches team members, and motivates them to perform at their best. To excel in project management, one must possess diverse skills. Effective project managers should demonstrate exceptional organisational capabilities, swiftly solve problems and communicate clearly. This course focuses on imparting project management concepts and techniques, offering learners a structured approach to understanding these concepts step-by-step.

ABOUT INSTRUCTOR :

Prof. Ramesh Anbanandam has supply chain management and transportation expertise. Currently serving as Professor in the Department of Management Studies at the Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India, Prof. Ramesh's research interests revolve around transportation and supply chain management. He specialises in sustainable multimodal transportation logistics, humanitarian supply chains, healthcare supply chains and waste management, and the adoption of electric vehicles in India. Furthermore, he is enthusiastic about exploring the potential applications of artificial intelligence (AI) in transportation and supply chain management problems.

Prof. Ramesh is an accomplished teacher who has taught several courses in the management field, including supply chain management, data analytics, project management, and operations research. He is also passionate about teaching the latest topics, such as IoT applications in management, circular supply chains for sustainability, and data science and big analytics. He has supervised 10 PhD theses and 13 M. Tech projects and has been recognised for his outstanding teaching abilities with the Outstanding Teacher Award in 2022 from IIT Roorkee.

Prof. Ramesh has edited three books, including "Flexibility, Innovation, and Sustainable Business" in collaboration with Santosh Rangnekar, which was published by Springer in 2022. He has also edited "Lean and Green Manufacturing: Towards Eco-Efficiency and Business Performance" with M. Kallian, H. Vimal, and A. Agrawal, which was published by Springer in 2021. Additionally, he was a co-author of the Indian edition of the book "Project Management: A Strategic Managerial Approach" published by Wiley in 2022.

Besides his academic and teaching responsibilities, Prof. Ramesh has taken up several administrative responsibilities at the institute and department levels. He also organised two major conferences, the 18th Annual International Conference of the Society of Operations Management in 2014 and the 19th Global Conference on Flexible Systems Management, GLOGIFT-19, in 2019.

Prof. Ramesh is a Life Member of four professional societies, including the Indian Institution of Industrial Engineers, the Society of Operations Management, the Global Institute of Flexible Systems Management, and the Indian Society for Technical Education. He has received several accolades for his contributions to the field of management, including the Award for Excellence in PhD Research in 2020 from IIT Roorkee and the Literati Award for Excellence in 2011 and 2016 under the "Highly Commended Research Paper in the Year" category for his papers published in Emerald publications. He is currently the Associate Editor for the International Journal of Revenue Management, published by Inderscience Publishers and listed under the ABDC-C category.

COURSE PLAN :

Week 1:

Part-I Project Initiation

Lecture 1 - Introduction to project management - I

Lecture 2 - Introduction to project management -II

Lecture 3 - Agile project management

Lecture 4 - Project strategy and selection overview

Lecture 5 - Project selection models

Week 2:

Part-I Project Initiation

Lecture 6 - Project manager

Lecture 7 - Attributes of Effective Project Manager

Lecture 8 - Managing for stakeholders

Lecture 9 - Resolving Conflicts

Lecture 10 - Negotiation

Week 3:

Part-I Project Initiation

Lecture 11 - Project in the organization structure

Lecture 12 - Human factors and the project team

Part-II Project Planning

Lecture 13 - Traditional project activity planning

Lecture 14 - Agile project planning, Project charter

Lecture 15 - Coordination through integration management

Week 4:

Part-II Project Planning

Lecture 16 - Project feasibility analysis,

Lecture 17 - Estimating project budgets

Lecture 18 - Project risk management

Lecture 19 - Quantitative risk assessment methodologies

Lecture 20 - Critical path method (CPM)

Week 5:

Part-II Project Planning

Lecture 21 - Programme evaluation and review technique (PERT)

Lecture 22 - Risk analysis with simulation for scheduling

Lecture 23 - Gantt Chart, Scheduling with scrum

Lecture 24 - Crashing a project

Lecture 25 - Resource loading

Week 6:

Part-II Project Planning

Lecture 26 - Resource levelling, Case study on Statue of Unity

Lecture 27 - Goldratt's critical chain

Part-III Project Execution

Lecture 28 - Planning-monitoring-controlling cycle

Lecture 29 - Earned value analysis

Lecture 30 - Agile tools for tracking project

Week 7:

Part-III Project Execution

Lecture 31 - Three types of project-controlling

Lecture 32 - Control of change scope and scope creep

Lecture 33 - Project audit,

Lecture 34 - Essentials of an audit/evaluation

Lecture 35 - When to close a project

Week 8:

Part-III Project Execution

Lecture 36 - Benefits realisation, Case study on the success of Chandrayan-3

Part-IV IT for Project Management

Lecture 37 - Software for project management

Lecture 38 - Demo on project management software

Lecture 39 - Simulations software for project management

Lecture 40 - Course Summary

MOOC APPROVAL PROPOSAL

KTU Course Name	Advanced Cryptography
Course Category	Honours
Course Code	HNCST709
MOOC Course Name	Cryptography and Network Security
NPTEL Subject ID	106105162
Course ID	noc26-cs57
Course URL	https://nptel.ac.in/courses/106105162
COORDINATORS	Prof. Sourav; Mukhopadhyay
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER 7

Advanced Cryptography

Course Code	HNCS709	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To equip the learner in the theoretical foundations of quantum mechanics and quantum algorithms, such as Shor's and Grover's, in order to evaluate their destabilizing impact on classical public-key cryptography and understand the principles of quantum key distribution.
2. To investigate the mathematical hard problems and design principles underlying post-quantum cryptographic families—specifically hash-based, code-based, lattice-based, and multivariate systems—to develop secure alternatives resilient against quantum adversaries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quantum Computing-Introduction, Quantum mechanics Shor's Factoring Algorithm-Factoring, Reduction from factoring to period-finding, Shor's period-finding algorithm, Continued fractions Grover's Search Algorithm, Quantum Cryptography-Saving cryptography from Shor, Quantum key distribution, Reduced density matrices and the Schmidt decomposition, The impossibility of perfect bit commitment, More quantum cryptography (Quantum Computing: Lecture Notes- Ronald de Wolf)	9
2	Introduction to post-quantum cryptography- Is cryptography dead?, A taste of post-quantum cryptography, Challenges in post-quantum cryptography, Comparison to quantum cryptography Classical Cryptography and Quantum computing, The computational model, The quantum Fourier transform, The hidden subgroup problem	9
3	Hash-based Digital Signature Schemes-Hash based one-time signature schemes, Merkle's tree authentication scheme, One-time key-pair generation using an PRNG, Authentication path computation, Tree chaining, Distributed signature generation, Security of the Merkle Signature Scheme Code-based cryptography-Introduction, Cryptosystems, The security	9

	of computing syndromes as one-way function, Codes and structures	
4	Lattice-based Cryptography-Introduction, Preliminaries, Finding Short Vectors in Random q -ary Lattices, Hash Functions, Public Key Encryption Schemes, Digital Signature Schemes Multivariate Public Key Cryptography-Introduction, The Basics of Multivariate PKCs, Examples of Multivariate PKCs, Basic Constructions and Variations, Standard Attacks (module 2-4 -Post-Quantum Cryptography by Daniel J. Bernstein, Johannes Buchmann, Erik Dahmen)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop problem-solving and analytical skills in quantum computation	K3
CO2	Describe the foundations and goals of post-quantum cryptography	K2
CO3	Apply authentication mechanisms in hash-based systems and understand the foundations of code-based cryptography	K3
CO4	Apply lattice-based cryptographic constructions and Multivariate Public Key Cryptography for data security	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2								2
CO2	3										2
CO3	3	2	2								2
CO4	3	2	2								2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Post-Quantum Cryptography	Daniel J. Bernstein · Johannes Buchmann Erik Dahmen	Springer	1/e, 2009
2	Quantum Computing	Ronald de Wolf	QuSoft, CWI and University of Amsterdam	I/e, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1				
2				
3				

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	
2	
3	
4	

MODEL QUESTION PAPER
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SEVENTH SEMESTER B. TECH HONOURS DEGREE EXAMINATION, MONTH AND YEAR
Course Code:



CRYPTOGRAPHY AND NETWORK SECURITY

PROF. SOURAV MUKHOPADHYAY

Department of Computer Science and Engineering
IIT Kharagpur

INDUSTRIES APPLICABLE TO : Stratign FZE, Dubai(UAE), SAG, DRDO, ISRO, WESEE, NTRO.

COURSE OUTLINE :

The aim of this course is to introduce the student to the areas of cryptography and cryptanalysis. This course develops a basic understanding of the algorithms used to protect users online and to understand some of the design choices behind these algorithms. Our aim is to develop a workable knowledge of the mathematics used in cryptology in this course. The course emphasizes to give a basic understanding of previous attacks on cryptosystems with the aim of preventing future attacks. A wide variety of basic cryptographic primitives will be discussed along with recent developments in some advanced topics like identity-based encryption, attribute-based encryption, functional encryption, two-party/multi-party computation, bitcoin and crypto-currency and postquantum cryptography. The cryptanalysis part will help us understanding challenges for cybersecurity that includes network security, data security, mobile security, cloud security and endpoint security.

ABOUT INSTRUCTOR :

Sourav Mukhopadhyay is an Associate Professor, Department of Mathematics at Indian Institute of Technology Kharagpur. He has completed his B.Sc (Honours in Mathematics) in 1997 from University of Calcutta, India. He has done M.Stat (in statistics) and M.Tech (in computer science) from Indian Statistical Institute, India, in 1999 and 2001 respectively. He worked with Cryptology Research Group at Indian Statistical Institute as a PhD student and received his Ph.D. degree in Computer Science from there in 2007. He was a Research Assistant at the Computer Science department of School of Computing, National University of Singapore (NUS). He visited Inria Rocquencourt, project CODES, France and worked as a post-doctoral research fellows at the School of Computer Engineering, Nanyang Technological University (NTU), Singapore. He was a post-doctoral research fellows and a part time Lecturer with School of Electronic Engineering, Dublin City University (DCU), Ireland.

COURSE PLAN :

Week 1: Introduction to cryptography, Classical Cryptosystem, Block Cipher.

Week 2: Data Encryption Standard (DES), Triple DES, Modes of Operation, Stream Cipher.

Week 3: LFSR based Stream Cipher, Mathematical background, Abstract algebra, Number Theory.

Week 4: Modular Inverse, Extended Euclid Algorithm, Fermats Little Theorem, Euler Phi-Function, Eulers theorem.

Week 5: Advanced Encryption Standard (AES), Introduction to Public Key Cryptosystem, Diffie-Hellman Key Exchange

Week 6: Primarily Testing, ElGamal Cryptosystem, Elliptic Curve over the Reals, Elliptic curve Modulo a Prime.

Week 7: Generalized ElGamal Public Key Cryptosystem, Rabin Cryptosystem.

Week 8: Message Authentication, Digital Signature, Key Management, Key Exchange, Hash Function.

Week 9: Cryptographic Hash Function, Secure Hash Algorithm (SHA), Digital Signature Standard (DSS).

Week 10: Cryptanalysis, Time-Memory Trade-off Attack, Differential and Linear Cryptanalysis.

Week 11: Cryptanalysis on Stream Cipher, Modern Stream Ciphers, Shamirs secret sharing and BE, Identity-based Encryption (IBE), Attribute-based Encryption (ABE).

Week 12: Side-channel attack, The Secure Sockets Layer (SSL), Pretty Good Privacy (PGP), Introduction to Quantum Cryptography,, Blockchain, Bitcoin and Cryptocurrency.

MOOC APPROVAL PROPOSAL

KTU Course Name	Object Oriented Design using UML
Course Category	Honours
Course Code	HNCST509
MOOC Course Name	Object Oriented System Development using UML, Java And Patterns
NPTEL Subject ID	106105224
Course ID	noc26-cs46
Course URL	https://nptel.ac.in/courses/106105224
COORDINATORS	Prof. Rajib Mall
Offering Institute	IIT Kharagpur
Duration	12 weeks
Platform	NPTEL

SEMESTER 5

Object Oriented Design Using UML

Course Code	HN C ST509	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To master the fundamental principles of object-oriented methodology—including abstraction, encapsulation, and inheritance—and apply analysis techniques to model the static and dynamic behavior of software systems.
2. To implement rigorous system and object design strategies and utilize the Unified Modeling Language (UML) along with standard design patterns to construct robust, scalable software architectures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Object Oriented Development - Modeling Concepts–Object Oriented Methodology. Object Oriented Themes - Abstraction - Encapsulation - Combining Data and Behavior - Sharing - Emphasis on Object Structure. Object Oriented Models. Object modeling: Objects and Classes, Links and Associations, Advanced links and Association Concepts, Generalization and Inheritance, Grouping Constructs, A Sample Object Model.	11
2	Dynamic modeling: Events and States, Operations, Nested state diagrams, Concurrency, Advanced Dynamic Modeling Concepts, A sample Dynamic Model, Relationship of Object and Dynamic models. Functional modeling: Functional models, Data Flow Diagrams, Specifying Operations, Constraints, A sample Functional Model. Analysis: Object Modeling - Identifying Object Classes - Preparing a Data Dictionary - Identifying Associations. Dynamic Modeling - Preparing a Scenario - Interface Format - Identifying Event - Building a State Diagram. Functional Modeling - Identifying input and Output Values - Building Data Flow Diagram-Describing Functions.	11

3	System Design: Breaking System into Subsystems, Identifying Concurrency, Allocating Subsystems to Processors and Tasks, Managing Data Stores, Handling of Global Resources, Common Architectural Framework. Object Design: Overview of Object design, Combining the three models, Designing algorithms, Design optimization, Implementation of control, Adjustment of inheritance, Design of association, Object representation, Physical packaging. Documenting design decisions - Comparison of methodologies.	11
4	Unified Modeling Language (UML)-UML introduction & benefits-Different types of UML diagrams- Behavioral. Diagrams-Activity Diagram - Use Case Diagram - State Machine Diagram - Sequence Diagram , structural diagrams.- Class Diagram-Object Diagram-Component Diagram-Composite Structure Diagram-Deployment Diagram-Package Diagram. UML tools and their needs. Design Patterns-Creational - Structural – Behavioral.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop object models that represent the static structure of a real-world system by utilizing concepts such as classes, associations, generalization, and inheritance.	Apply
CO2	Develop dynamic and functional models using state transition diagrams and data flow diagrams to capture system events, concurrency, and operational requirements.	Apply
CO3	Translate analysis models into detailed system and object designs by defining subsystems, optimizing algorithms, and managing data storage strategies.	Apply
CO4	Solve real time problems using various Modeling concepts for managing projects in multidisciplinary environments	Apply

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								3
CO2	3	3	3	3							3
CO3	3	3	3	3							3
CO4	3	3	3	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Object Oriented Modeling and Design	James Rumbaugh	Prentice Hall India	1/e
2	Object Oriented Analysis and Design with Applications	Grady Booch	Pearson Education Asia References	3/e
3	UML Distilled: A Brief Guide to the Standard Object Modeling Language	Martin Fowler	Addison-Wesley Professional	3/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Object Oriented Software Engineering	Ivan Jacobson	Pearson Education Asia.	3/e
2	Object Oriented Software	Berno Bruegge, Allen H.	Pearson Education	3/e



OBJECT ORIENTED SYSTEM DEVELOPMENT USING UML, JAVA AND PATTERNS

PROF. RAJIB MALL

Department of Computer Science and Engineering
IIT Kharagpur

PRE-REQUISITES : Programming Using Java, Software Engineering

INTENDED AUDIENCE : CSE, IT

COURSE OUTLINE :

Object-oriented software development has become very popular. Also, UML has been accepted as the standard design language. We discuss use of UML to arrive at a design solution. Skeletal java code generation from UML diagrams will be discussed. Design patterns are reusable solutions. These are good solutions to typical programming problems, that can be understood and applied in a specific design situation to improve the overall design and reduce design iterations.

ABOUT INSTRUCTOR :

Prof. Rajib Mall is Professor, Department of Computer Science and Engineering, Indian Institute of Technology Kharagpur, West Bengal. He has more than a two decades of teaching experience in the areas of real-time systems, program analysis and testing. He has written five text books and over 150 refereed research papers.

COURSE PLAN :

Week 1: Introduction

Week 2: Life Cycle Models for OO Development

Week 3: Use Case Diagram

Week 4: Class Diagram I

Week 5: Class Diagram II

Week 6: Sequence Diagram

Week 7: State chart diagram

Week 8: Design process

Week 9: Introduction to design patterns

Week 10: GRASP patterns

Week 11: GoF pattern I

Week 12: GoF Pattern II

MOOC APPROVAL PROPOSAL

KTU Course Name	Advanced Algorithms
Course Category	Honours
Course Code	HNCST609
MOOC Course Name	Design and Analysis of Algorithms
NPTEL Subject ID	106106131
Course ID	noc26-cs67
Course URL	https://nptel.ac.in/courses/106106131
COORDINATORS	Prof. Madhavan Mukund
Offering Institute	Chennai Mathematical Institute
Duration	8 weeks
Platform	SWAYAM

SEMESTER 6

Advanced Algorithms

Course Code	HNCST609	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:1:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCST502	Course Type	Theory

Course Objectives:

1. To introduce advanced algorithmic techniques and theoretical tools for analyzing complex problems.
2. To enable students to design and evaluate efficient solutions using dynamic, greedy, randomized, and approximation strategies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Advanced recurrence forms, Akra-Bazzi Theorem, Dynamic Programming - Longest Common Subsequences, Bellman Ford algorithm, Backtracking - Subset Sum, Hamiltonian Path (concept only), Branch and Bound - Knapsack problem, Greedy Method - Huffman codes, Matroids. Space-Bounded Computations - Basic concepts of L & NL, Space hierarchy (introductory), Savitch's Theorem (concept only).	11
2	Sorting Networks - Comparison Networks, Zero-One Principle, Bitonic Sorting Network - structure & analysis, Merging Networks, Batcher's Odd-Even Mergesort Network, Complexity of Sorting Networks (depth & size). String Matching Algorithms - The Naïve Pattern Matching Algorithm, Rabin-Karp Algorithm - analysis using hashing, Finite Automata-based String Matching, Knuth -Morris-Pratt (KMP) Algorithm - prefix function, analysis.	11
3	Randomization - Basic Probability - indicator variables, inequalities & Bounds - Markov's Inequality, Chebyshev's Inequality, Chernoff Bound (applications only) - Universal Hashing - Expectations, Markov Chains and Random Walks, 2-SAT random walk, random walks on graphs (concept only), Applications of Randomized Algorithms.	11

4	Approximation Algorithms - Approximation Algorithms for NP -Hard Problems - Approximation Algorithms for the Vector cover problem, Traveling Salesman Problem - Knapsack Problem, Algorithms for Solving Nonlinear Equations - Bisection Method - Method of False Position - Newton's Method.	11
---	---	----

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain advanced algorithmic concepts such as recurrence solving, sorting networks, randomized bounds and space-bounded computations.	K2
CO2	Summarize key algorithmic strategies including dynamic programming, greedy methods, backtracking and approximation techniques.	K2
CO3	Apply dynamic programming, greedy and randomized methods	K3

	to solve problems such as LCS, Bellman-Ford, KMP and hashing-related tasks.	
CO4	Apply approximation algorithms and numerical methods to obtain solutions for NP-hard problems and nonlinear equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							2
CO2	3	3	3	2							2
CO3	3	3	3	2	2						2
CO4	3	3	3	2							2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	MIT Press	4/e, 2022
2	Randomized Algorithms	Rajeev Motwani and Prabhakar Raghavan	Cambridge University Press	1/e, 2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Algorithms	Robert Sedgewick and Kevin Wayne	Addison-Wesley	4E, 2011
2	Algorithm Design	Jon Kleinberg & Éva Tardos	Addison-Wesley	1E, 2006
3	Introduction to the Theory of Computation	Michael Sipser	Cengage Learning	3E, 2012
4	Approximation Algorithms	Vijay V. Vazirani	Springer-Verlag Berlin Heidelberg	1E, 2001



DESIGN AND ANALYSIS OF ALGORITHMS

PROF. MADHAVAN MUKUND

Department of Computer Science and Engineering
Chennai Mathematical Institute

INTENDED AUDIENCE: Students in BE/BTech Computer Science, 2nd/3rd year.

PRE-REQUISITES: Exposure to introductory courses on programming and data structures.

INDUSTRY SUPPORT: This course should be of value to any company working in the area of software services and products.

COURSE OUTLINE:

This course will cover basic concepts in the design and analysis of algorithms.

- Asymptotic complexity, $O()$ notation
- Sorting and search
- Algorithms on graphs: exploration, connectivity, shortest paths, directed acyclic graphs, spanning trees
- Design techniques: divide and conquer, greedy, dynamic programming
- Data structures: heaps, union of disjoint sets, search trees
- Intractability

ABOUT INSTRUCTOR :

Prof. Madhavan Mukund studied at IIT Bombay (BTech) and Aarhus University (PhD). He has been a faculty member at Chennai Mathematical Institute since 1992, where he is presently Professor and Director. His main research area is formal verification. He has active research collaborations within and outside India and serves on international conference programme committees and editorial boards of journals.

He has served as President of both the Indian Association for Research in Computing Science (IARCS) (2011-2017) and the ACM India Council (2016-2018). He has been the National Coordinator of the Indian Computing Olympiad since 2002. He served as the Executive Director of the International Olympiad in Informatics from 2011-2014.

In addition to the NPTEL MOOC programme, he has been involved in organizing IARCS Instructional Courses for college teachers. He is a member of ACM India's Education Committee. He has contributed lectures on algorithms to the Massively Empowered Classroom (MEC) project of Microsoft Research and the QEEE programme of MHRD.

COURSE PLAN:

Week 1:

Module 1: Introduction

Module 2: Examples and motivation

Module 3: Examples and motivation

Module 4: Asymptotic complexity: informal concepts

Module 5: Asymptotic complexity: formal notation

Module 6: Asymptotic complexity: examples

Assignments MCQ/Fill in blanks (unique answer)

Week 2

Module 1: Searching in list: binary search

Module 2: Sorting: insertion sort

Module 3: Sorting: selection sort

Module 4: Sorting: merge sort

Module 5: Sorting: quicksort

Module 6: Sorting: stability and other issues

Assignments MCQ/Fill in blanks, programming assignment

Week 3

Module 1: Graphs: Motivation

Module 2: Graph exploration: BFS

Module 3: Graph exploration: DFS

Module 4: DFS numbering and applications

Module 5: Directed acyclic graphs

Module 6: Directed acyclic graphs

Assignments MCQ/Fill in blanks, programming assignment

Week 4

Module 1: Shortest paths: unweighted and weighted

Module 2: Single source shortest paths: Dijkstra

Module 3: Single source shortest paths: Dijkstra

Module 4: Minimum cost spanning trees: Prim's algorithm

Module 5: Minimum cost spanning trees: Kruskal's Algorithm

Module 6: Union-Find data structure

Assignments MCQ/Fill in blanks, programming assignment

Week 5

Module 1: Divide and conquer: counting inversions

Module 2: Divide and conquer: nearest pair of points

Module 3: Priority queues, heaps

Module 4: Priority queues, heaps

Module 5: Dijkstra/Prims revisited using heaps

Module 6: Search Trees: Introduction

Assignments MCQ/Fill in blanks, programming assignment

Week 6

Module 1: Search Trees: Traversals, insertions, deletions

Module 2: Search Trees: Balancing

Module 3: Greedy : Interval scheduling

Module 4: Greedy : Proof strategies

Module 5: Greedy : Huffman coding

Module 6: Dynamic Programming: weighted interval scheduling

Assignments MCQ/Fill in blanks, programming assignment

Week 7

Module 1: Dynamic Programming: memoization

Module 2: Dynamic Programming: edit distance

Module 3: Dynamic Programming: longest ascending subsequence

Module 4: Dynamic Programming: matrix multiplication

Module 5: Dynamic Programming: shortest paths: Bellman Ford

Module 6: Dynamic Programming: shortest paths: Floyd Warshall

Assignments MCQ/Fill in blanks, programming assignment

Week 8

Module 1: Intractability: NP completeness

Module 2: Intractability: reductions

Module 3: Intractability: examples

Module 4: Intractability: more examples

Module 5: Misc topics

Module 6: Misc topics

Assignments MCQ/Fill in blanks

MOOC APPROVAL PROPOSAL

KTU Course Name	Web Programming
Course Category	Program Elective 4
Course Code	PECST742
MOOC Course Name	Web Technology
Course URL	https://onlinecourses.swayam2.ac.in/nou24_cs09/preview
COORDINATORS	Dr. Ashutosh Kumar Bhatt
Offering Institute	Uttarakhand Open University, Haldwani
Duration	12 weeks
Platform	SWAYAM

SEMESTER S7

WEB PROGRAMMING

(Common to CS/CA/CM/CD/CR/AD/AM)

Course Code	PECST742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/	Course Type	Theory

Course Objectives:

1. To equip students with the knowledge and skills required to create, style, and script web pages using HTML5, CSS, JavaScript, and related technologies.
2. To provide hands-on experience with modern web development tools and frameworks such as React, Node.js, JQuery, and databases, enabling students to design and build dynamic, responsive, and interactive web applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Creating Web Page using HTML5 - Introduction, First HTML5 example, Headings, Linking, Images, Special Characters and Horizontal Rules, Lists, Tables, Forms, Internal Linking, meta Elements, HTML5 Form input Types, Input and datalist Elements and autocomplete Attribute, Page-Structure Elements; Styling Web Page using CSS - Introduction, Inline Styles, Embedded Style Sheets, Linking External Style Sheets, Positioning Elements:, Absolute Positioning, z-index, Positioning Elements: Relative Positioning, span, Backgrounds, Element Dimensions, Box Model and Text Flow, Media Types and Media Queries, Drop-Down Menus; Extensible Markup Language - Introduction, XML Basics, Structuring Data, XML Namespaces, Document Type Definitions (DTDs), XML Vocabularies	9
2	Scripting language - Client-Side Scripting, Data Types, Conditionals, Loops, Arrays , Objects , Function Declarations vs. Function Expressions , Nested Functions , The Document Object Model (DOM) - Nodes and NodeLists, Document Object, Selection Methods, Element Node Object, Event Types Asynchronous JavaScript and XML - AJAX : Making Asynchronous	9

	Requests , Complete Control over AJAX , Cross-Origin Resource Sharing JavaScript library - jQuery - jQuery Foundations - Including jQuery, jQuery Selectors, Common Element Manipulations in jQuery, Event Handling in jQuery	
3	JavaScript runtime environment : Node.js - The Architecture of Node.js, Working with Node.js, Adding Express to Node.js; Server-side programming language : PHP - What Is Server-Side Development? Quick tour of PHP, Program Control , Functions , Arrays , Classes and Objects in PHP , Object-Oriented Design ; Rendering HTML : React - ReactJS Foundations : The Philosophy of React, What is a component? Built- in components, User-defined components - Types of components, Function Components, Differences between Function and Class Components	9
4	SPA – Basics, Angular JS; Working with databases - Databases and Web Development, SQL, Database APIs, Accessing MySQL in PHP; Web Application Design - Real World Web Software Design, Principle of Layering , Software Design Patterns in the Web Context, Testing; Web services - Overview of Web Services - SOAP Services, REST Services, An Example Web Service, Web server - hosting options	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop structured web pages with HTML5 and style them using CSS techniques, including positioning, media queries, and the box model.	K3
CO2	Write client-side scripts using JavaScript and utilize jQuery for DOM manipulation, event handling, and AJAX requests to create responsive and interactive user interfaces.	K3
CO3	Build and deploy server-side applications using Node.js, Express, and PHP, and integrate databases using SQL to store and retrieve data for dynamic content generation.	K3
CO4	Utilize React for building component-based single-page applications (SPAs), understanding the fundamental principles of component architecture, and leveraging AngularJS for web application development.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Web Development	Randy Connolly, Ricardo Hoar	Pearson	1/e, 2017
2	Building User Interfaces with ReactJS - An Approachable Guide	Chris Minnick	Wiley	1/e, 2022
3	Internet & World Wide Web - How to Program	Paul J. Deitel, Harvey M. Deitel, Abbey Deitel	Pearson	1/e, 2011
4	SPA Design and Architecture: Understanding Single Page Web Applications	Emmit Scott	Manning Publications	1/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Hand Book On Web Development : From Basics of HTML to JavaScript and PHP	Pritma Jashnani	Notion press	1/e, 2022
2	Advanced Web Development with React	Mohan Mehul	BPB	1/e, 2020
3	JavaScript Frameworks for Modern Web Development	Tim Ambler, Sufyan bin Uzayr, Nicholas Cloud	Apress	1/e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106222/
2	https://archive.nptel.ac.in/courses/106/106/106106156/

[Home](#) > [Courses](#) >

Web Technology

By Dr. Ashutosh Kumar Bhatt | Uttarakhand Open University, Haldwani

Learners enrolled: 10843


Web Technology refers to the many different tools and techniques that are utilized in the process of communication between different types of devices over the internet in Hindi/English Language. The main sections or components of Web Technology are World Wide Web (WWW), Web Browser, Web Server, Web Pages, and Web Development. It can be classified into two ways: Frontend and Backend Development. The part of a website where the user interacts directly is termed as front end. It is also referred to as the client-side of the application. Backend is the server side of a website. It is the part of the website that users cannot see and interact with. It is the part of software that does not come in direct contact with the users. It is used to arrange/store data. The frontend languages are HTML, CSS, JavaScript, AJAX, etc. and Backend Languages are PHP, Node.js, Python, Ruby, Java, JavaScript, C# etc.

Summary

Course Status :	Completed
Course Type :	Not Applicable
Language for course content :	English
Duration :	12 weeks
Category :	Computer Science and Engineering
Level :	Undergraduate/Postgraduate
Start Date :	01 Jan 2024
End Date :	30 Apr 2024
Enrollment Ends :	29 Feb 2024
Exam Date :	19 May 2024
	IST
Shift :	Shift-I

Note: This exam date is subject to change based on seat availability. You can check final exam date on your hall ticket.

Course layout

Week	Topic	Module
Week 1	MARKUP LANGUAGES Web Standards, The History of Markup Languages HTML, XML, XHTML	MARKUP LANGUAGES Web Standards, The History of Markup Languages
	WML-Wireless Markup Language Cascading Style Sheets (CSS) DHTML	HTML, XML, XHTML
	Client-Side and Server-Side Technologies Client-Side and Server-Side Technologies Basics of HTML	WML-Wireless Markup Language, Cascading Style Sheets (CSS)
	HTML Document Structure	DHTML and Client-Side and Server-Side Technologies

		Basics of HTML
		HTML Document Structure
Week 2	HTML, Introduction to Hyperlinks, Hyperlinks - Examples Links to the same document, HTML, Frames and Framesets Frame attributes Nested framesets, HTML Tables, Tables Attributes , Rows and Columns	HTML
		Introduction to Hyperlinks, Hyperlinks - Examples Links to the same document
		HTML, Frames
		Framesets
		Frame attributes Nested framesets
		HTML Tables, Tables Attributes , Rows and Columns
Week 3	Introduction to Forms The <FORM>, <INPUT> Tag Text input, CASCADING STYLE SHEETS (CSS) Inline Style sheet , Internal Style Sheet , External Style sheet, CASCADING STYLE SHEETS, Imported Style Sheet ,Cascading Rules , Order rules , Order rules	Introduction to Forms
		The <FORM>, <INPUT> Tag Text input
		CASCADING STYLE SHEETS (CSS) Inline Style sheet ,
		Internal Style Sheet , External Style sheet
		CASCADING STYLE SHEETS, Imported Style Sheet
		Cascading Rules , Order rules , Order rules
Week 4	CSS, Difference between ID and Classes , Anatomy of a style rule , Style Classes , Attribute Selector XHTML – eXtensible Hyper Text Markup Language , Need for XHTML XHTML – Base syntactic rules , XHTML – other syntactic rules , DTD – Document Type Definition	CSS, Difference between ID and Classes
		Anatomy of a style rule , Style Classes , Attribute Selector
		XHTML – eXtensible Hyper Text Markup Language
		Need for XHTML
		XHTML – Base syntactic rules , XHTML – other syntactic rules
		DTD – Document Type Definition
Week 5	XML ,XML Trees Namespaces , Defining XML Data Formats JAVASCRIPT AND HTML Document Object Model or DOM	XML
		XML Trees
		Namespaces
		Defining XML Data Formats
		JAVASCRIPT
		HTML
		Document Object Model or DOM
Week 6	JavaScript - Document Object Model or DOM , Types of DOM nodes JavaScript - Traversing the DOM tree , Working with DOM JAVASCRIPT AND EVENT HANDLING- JavaScript - Event Handlers	JavaScript - Document Object Model or DOM
		Types of DOM nodes
		JavaScript - Traversing the DOM tree
		Working with DOM
		JAVASCRIPT AND
		EVENT HANDLING
		JavaScript - Event Handlers
Week 7	JAVASCRIPT AND EVENT HANDLING- onClick event type , onSubmit event type	JAVASCRIPT AND EVENT HANDLING

	JAVASCRIPT AND EVENT HANDLING- onMouseOver and onMouseOut event JAVASCRIPT AND EVENT HANDLING- java script cookies , create cookies	onClicK event type
		onSubmit event type
		JAVASCRIPT AND EVENT HANDLING
		onMouseOver and onMouseOut event
		JAVASCRIPT AND EVENT HANDLING
		java script cookies , create cookies
Week 8	JAVASCRIPT AND EVENT HANDLING- javascript cookies , Store Cookies JAVA SERVER PAGES(JSP), Introduction to JSP , Comparison of Servlets with JSP Advantages of JSP , JSP engines , JSP Architecture	JAVASCRIPT AND EVENT HANDLING- javascript cookies
		Store Cookies
		JAVA SERVER PAGES(JSP)
		Introduction to JSP
		Comparison of Servlets with JSP
		Advantages of JSP
		JSP engines, JSP Architecture
Week 9	Life cycle of a JSP Page , Directory structure of JSP , Anatomy of a JSP page JSP Components and Tags JSP Creation of a small Web application using JSP	Life cycle of a JSP Page
		Directory structure of JSP
		Anatomy of a JSP page
		JSP Components
		JSP Tags
		JSP Creation
		A Small Web application using JSP
Week 10	JAVA SERVER PAGES(JSP) , JSP Elements – JSP Actions JSP Implicit Objects JSP - Exception Handling	JAVA SERVER PAGES(JSP) Basics
		JSP Elements –
		JSP Elements Examples
		JSP Actions
		JSP Actions Examples
		JSP Implicit Objects
		JSP - Exception Handling
Week 11	JSP Session Tracking JAVA SERVER PAGES (JSP) WITH JDBC- Basic steps to connect database , Install Mysql , Steps to run JSP using Tomcat server Creating a table using JSP example , Creating a table: database	JSP Session Tracking
		JAVA SERVER PAGES (JSP) WITH JDBC
		Install Mysql
		Steps to run JSP using Tomcat server
		Basic steps to connect database
		Creating a table using JSP

		example
		Creating a table: database
Week 12	JAVA SERVER PAGES (JSP) WITH JDBC- SELECT operation , INSERT operation , DELETE Operation JDBC – INTRODUCTION , JDBC , JDBC Architecture JDBC Drivers , Working with JDBC, JDBC - JDBC Object Classes JAVA SERVLET IMPLEMENTATION, Tomcat Installation JAVA SERVLET -Servlet Implementation JAVA SERVLET -The doGet method , The doPost method	JAVA SERVER PAGES (JSP) WITH JDBC- SELECT operation , INSERT operation , DELETE Operation
		JDBC – INTRODUCTION , JDBC , JDBC Architecture
		JDBC Drivers , Working with JDBC
		JDBC - JDBC Object Classes
		JAVA SERVLET IMPLEMENTATION, Tomcat Installation
		JAVA SERVLET -Servlet Implementation
		JAVA SERVLET -The doGet method , The doPost method

Books and references

- Powell. Thomas A., JavaScript: The Complete Reference
- Iemay. Laura, rafe colburn, jennifer kyrnin, Mastering HTML, CSS & JavaScript Web, BPB Publication, 2016
- Vishvajeet. Sisodia, Basic of Web Design, HTML, CSS3, Centrum Press, 2014.

Instructor bio



**Dr. Ashutosh
Kumar Bhatt**
Uttarakhand Open
University,
Haldwani

Dr. Ashutosh Kumar Bhatt is working in School of Computer Science and Information Technology, Uttarakhand Open University, Haldwani, Distt- Nainital (Uttarakhand). Dr. Bhatt has completed Ph. D. in Computer Science in 2009. His work area of research was Artificial Neural Network.

He has more than Seventeen years of teaching and research experience in various organizations of repute for PG & UG courses of Computer Science & IT.

He is also associated with many renowned National/ International Journals as Lead Guest Editor/reviewer/editorial board member. More than 12 SCIE/ESCI and more than 55 Scopus indexed research publications are credited to him in reputed National/International Journals.

He is a life member of CSI(Computer Society of India) and member of IEEE. He had served as State Student Coordinator of Region I Uttarakhand of CSI (for three years) and also served as Secretary of Uttarakhand ACM Professional Chapter.

He is awarded by the significant contribution award by CSI for the year 2012-13 and Distinguished Professor award by CSI Mumbai Chapter.

He has published and edited number of books e.g. C# Programming using Dot Net Framework, Technology Enabled Learning, Introduction to Computers with basics of Programming, Artificial Intelligence for Societal Development and Global Well-Being etc.

He had worked in a project entitled “Automated Analysis for Quality Assessment of Apples using Artificial Neural Network”, under DST New Delhi. He had also worked in the project entitled as “Automated IoT Based Smart Water Quality Assessment System” “Medicinal Knowledge Portal Exclusively for Uttarakhand